

Fragmentomic Liquid Biopsy Enables Early Breast Cancer Detection Molecular Subtyping and Lymph Node Assessment

Corresponding Author: Dr Chao Ni

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The manuscript by Zhu et al. describes classification models based on cfDNA fragmentomics for the early detection of breast cancer, molecular subtyping, and identification of disease metastasis, which have potential for improving the clinical management and treatment of breast cancer patients. Generally, the study is well-designed, supported by a large sample size, and includes valuable validation samples, such as blood specimens from patients with benign lesions confirmed by imaging diagnosis. The results appear to be robust and interpretable. However, the manuscript lacks sufficient background explanation regarding the rationale of the study design, and the comparisons with existing competitive methods are not adequately addressed, which should be strengthened to improve the overall clarity and contextual significance of the work.

Point 1

The authors report that their model achieved high performance, with an AUC > 0.93 for breast cancer detection, demonstrating a sensitivity of 75% at a specificity of approximately 85%. Presenting a table with multiple sensitivity and specificity pairs would enhance clarity. Furthermore, discussing which of these two metrics—sensitivity or specificity—is more critical in the context of real-world breast cancer detection and clinical decision-making would be helpful for clinicians to intuitively understand the value of this technique.

Point 2

Although the Discussion section (from line 455 onward) provides relevant interpretations, the Introduction generally lacks sufficient explanation regarding the overall study design and comparisons with other technologies. For example, the use of transcription factor (TF) motifs and repeat elements in the model should be clearly stated in the Introduction, along with appropriate background information. In addition, some references (e.g., references #4 and #5) do not appear to be the most appropriate for supporting the claims made. Given the growing number of liquid biopsy technologies for cancer detection, a more thorough comparison with existing approaches should be presented in the Introduction. Finally, the performance of the model in breast cancer detection and subtyping would be more informative if compared directly with that of other existing methods using a summary table (e.g. <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2023.1127086/full>, <https://www.nature.com/articles/s41523-021-00316-7>)

Point 3

At line 194, the authors describe filtering transcription factors (TFs) enriched in breast cancer using a breast cancer cell line. It would be helpful to include a discussion on how incorporating TFs enriched in benign conditions might influence model performance.

Point 4

In Figure 3K, the model correctly identified 25 out of 26 samples. Given the importance of early-stage detection—both in this dataset and in the broader clinical context—it would be valuable to present the model's performance specifically on early-stage samples, or stratified by stage. Also, while the 25/26 result in Figure 3K suggests outstanding early detection performance, a discussion comparing this result to the model's performance on the broader dataset would provide important context and help assess the generalizability of the approach.

Point 5

Regarding Figure 3B, it would be helpful to present the individual performance of each modality (DELF, end-motif, and repeat elements) at least in the Supplementary Materials. This would allow readers to better assess the relative significance of each modality and their respective contributions to the performance of the combined model.

Point 6

The authors highlight the advantage of their approach as being suitable for ultra-low coverage settings. To support this claim, it would be helpful to present the coverage distribution across samples. As the authors note, lower coverage is indeed advantageous in clinical situations due to constraints in blood volume and cost. Therefore, it would strengthen the manuscript to include an analysis of breast cancer detection performance across different coverage levels or by down-sampling samples to low coverages (0.1~1X). If feasible, similar analyses for the other models presented in the manuscript would also be valuable. Additionally, a discussion explaining why low-pass sequencing seems to be required.

Point 7

The authors claim that alternative technologies require high costs. To support this point, it would be helpful to include a comparison of the authors method to other existing methods regarding sequencing or assay costs.

Point 8

Figure 2A appears to show less centralized plots across samples compared to DELFI. Even in the healthy group, the patterns seem less centralized compared to the DELFI healthy reference. It would be helpful for the authors to provide a rationale or explanation for this observation.

Point 9

At lines 187–189, the manuscript suddenly shifts to discussing transcription factors (TFs), which may be confusing readers with the rationale for incorporating TF features. Providing additional background or a brief explanation would improve readers understanding. Similarly, at line 398, it is hard to get reasons why RNA-seq was chosen for downstream analysis, especially given the availability of other molecular data types such as methylation, SNVs, and CNVs. Although the rationale becomes somewhat clearer later in the Discussion (starting at line 478), the authors should consider introducing the reasoning earlier in the manuscript. A brief explanation of why xCell and ESTIMATE were selected—and how they support the study objectives—would also be helpful.

Point 10

In the manuscript showed the detection of lymph node metastasis. It would be helpful for the authors to clarify whether lymph node metastasis could be detected within a unified model along with other types of metastasis (Figure 4). If this is not feasible, a discussion explaining the limitations or reasons for separating these models would strengthen the manuscript.

Point 11

In the Discussion section (line 436), the authors mention the limitations of bisulfite-based methods. However, more recent approaches such as enzymatic C-to-T conversion and Oxford Nanopore Technologies (ONT)-based methylation profiling offer viable alternatives that may mitigate such limitations. The authors should provide more precise, performance-based evidence to support their claims. Furthermore, performance comparisons with existing studies—ideally in tabular format—would enhance the manuscript. In particular, at line 464, the authors compare their model to studies in other cancer types. It would be more clear to include direct comparisons with recent liquid biopsy studies of breast cancer.

Point 12

At line 438, the authors refer to their approach as “AI-based.” However, given the methodological context, it may be more appropriate to describe it as “machine learning-based,” as the use of “AI” could be considered too advanced or potentially misleading in this case.

Point 13

At line 457, the authors mention using a tumor score, but what is the corresponding limit of detection in terms of variant allele frequency (VAF)? A brief discussion on this point would clarify the model’s sensitivity.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The manuscript introduces a new multi-feature cfDNA-based Breast cancer detection method. The method is trained and tested on a cohort of 503 cases and 289 controls and shows good classification performance. Additionally, they train a ML model that can perform molecular subtyping using the same features. The results are pretty good and of interest to people in the field, but the methods section is lacking details about the features and how the method was trained.

You use three different kinds of fragmentomics features. How many features do you end up with of each kind? Have you tested which kind of fragmentomics features contribute the most to the performance of the method?

Describing the motif analysis, you write that the 6-bp motifs are centered around the breakpoint. Is this also the case for the 4bp motifs, or are they only the end-motif?

If you count motifs in both ends of the fragments, do you use the reverse complement of one of the ends so that the motifs are comparable?

There are no details about the different ML methods that are trained provided in the manuscript. I looked at the linked GitHub repo, but as far as I can see, that repo only contains code for plotting. You should both add more details about the parameters and implementations of the methods to the manuscript and add the code to GitHub to make the details regarding the training and testing of the models available.

For instance, is it a linear SVM you test, or do you use a kernel? Do you use cross-validation to fit any meta-parameters for any of the ML methods, or do you use default parameters? You end up using a Stacked Ensemble (I would write out the abbreviation the first time you use it), but that is just a method for combining other ML models, and you do not say anything about which models you are combining (is it the 5 other mentioned models?).

The method is called TuFEst (Tumor Fraction Estimator), but the manuscript only reports on classification performance; there are no plots showing an estimated tumor fraction compared to some independent estimate of the tumor fraction in the cfDNA samples.

(Remarks on code availability)

As far as I can see, that repo only contains code for plotting. I could not find code for training the different models mentioned in the manuscript.

Reviewer #4

(Remarks to the Author)

This is a well written manuscript demonstrating the power of cfDNA fragments to detect breast cancer as well as determine subtype and lymph node status.

It was a well conducted study using both benign and malignant breast tissue.

I think that this study will be of interest to researchers and clinicians involved in early breast cancer detections and treatment.

I have very minor concerns:

1. I could not locate Table 1 cited on line 209.
2. there is no discussion of tumors which are triple positive, whether they were included or excluded in the analyses.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Dear Editor,

I have received and carefully reviewed the authors' responses to reviewers comments by thoroughly addressing the points with appropriate justification and supporting results. Based on these revisions, the manuscript is appropriate for publication in Nature Communications and will provide valuable insights to researchers in the relevant field.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The inclusion of details about the model training as well as the feature importance analysis is a clear improvement. But there are two minor points I still feel need to be addressed:

1) The explanation of the 4 bp end motifs and 6bp breakpoint motifs in the manuscript remains cryptic. The clarification you wrote in the response to my comment is not reflected in the manuscript. In the graphical abstract, the 4bp and 6bp motifs are drawn the same way, and in the text, you do not distinguish between end motifs and breakpoint motifs. In line 252 you refer to them the same way: "encompassing end motifs (4/6 bp breakpoint signatures)". And line 285: "particular 4 bp (e.g., AAGC, AGGC) and 6 bp (e.g., CTTGTT, TGGGAC) end motifs". And line 348: "patterns in breakpoint end motifs". And line 641: "4 bp and 6 bp end motif frequencies".

2) I am surprised to see that you include so many Repetitive Element features, not just the main classes, but many subclasses. I can understand why Alu elements could be a relevant feature if they are actively transposing in the tumor tissue, but it is only a subset of the Alu subfamilies that can still be active. Furthermore, the results in Supp. Figure 2D shows that the main difference between benign and malignant samples is that there are more Alu reads in a subset of the benign

samples. What could explain this? The figure text conflicts with the legend. Line 1065: “breast malignancy (blue) and benign controls (pink)” but the legends specify the colors the other way around. Which is correct?

(Remarks on code availability)

It is not reproducible.

There is no README and no description of the input files (not included). There are some comments in the code but they are all in chinese.

Reviewer #4

(Remarks to the Author)

the authors have addressed all of my comments in this revised version

(Remarks on code availability)

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Point-by-point response

Reviewer #1 (Remarks to the Author):

The manuscript by Zhu et al. presents classification models utilizing cfDNA fragmentomics for the early detection of breast cancer, molecular subtyping, and identification of metastasis disease, showing promise for enhancing clinical management and treatment of breast cancer patients. Overall, the study is well-designed, with a large sample size and valuable validation cohorts—including blood samples from patients with imaging-confirmed benign lesions. The findings appear robust and interpretable.

However, the manuscript lacks sufficient background explanation regarding the rationale of the study design, and the comparisons with existing competitive methods are not adequately addressed, which should be strengthened to improve the overall clarity and contextual significance of the work.

Point 1

The authors report that their model achieved high performance, with an AUC > 0.93 for breast cancer detection, demonstrating a sensitivity of 75% at a specificity of approximately 85%. Presenting a table with multiple sensitivity and specificity pairs would enhance clarity. Furthermore, discussing which of these two metrics—sensitivity or specificity—is more critical in the context of real-world breast cancer detection and clinical decision-making would be helpful for clinicians to intuitively understand the value of this technique.

Response: We sincerely thank the reviewer for this constructive suggestion. In response, we have added a dedicated table summarizing multiple sensitivity–specificity operating points for TuFEst across both the training and validation cohorts, derived from different machine learning models. This table has been included as **Supplementary Table 2** in

the revised manuscript.

Regarding the relative importance of sensitivity versus specificity in real-world breast cancer early detection, we have expanded the discussion as follows (Lines 484–494 in the revised version):

“In real-world early detection, the optimal balance between sensitivity and specificity must be tailored to the clinical context. For population-level screening, particularly among Asian populations with dense breast tissue and suboptimal mammographic performance, higher sensitivity is essential to minimize the risk of missed early-stage cancers. Since our cfDNA fragmentomics approach is intended to complement routine ultrasound at the initial diagnostic stage, adopting a higher-sensitivity threshold supports timely referral to MRI or biopsy and reduces the likelihood of false-negative outcomes. At the same time, maintaining reasonable specificity is crucial to avoid unnecessary imaging or invasive procedures. In this regard, the TuFEst model achieves a clinically practical balance (specificity: 0.783 in training database; 0.869 in external validation database), enabling flexible integration across diverse healthcare settings and diagnostic workflows.”

Point 2

Although the Discussion section (from line 455 onward) provides relevant interpretations, the Introduction generally lacks sufficient explanation regarding the overall study design and comparisons with other technologies. For example, the use of transcription factor (TF) motifs and repeat elements in the model should be clearly stated in the Introduction, along with appropriate background information. In addition, some references (e.g., references #4 and #5) do not appear to be the most appropriate for supporting the claims made. Given the growing number of liquid biopsy technologies for cancer detection, a more thorough comparison with existing approaches should be presented in the Introduction. Finally, the performance of the model in breast cancer detection and subtyping would be more informative if compared directly with that of other existing methods using a summary table (e.g. <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2023.1127086/ful>

l, <https://www.nature.com/articles/s41523-021-00316-7>)

Response: We thank the reviewer for these excellent suggestions. Firstly, we expanded the Introduction to briefly outline our overall study design and to clarify the rationale for incorporating multiple fragmentomic modalities, as described in Lines 144–150 of the revised manuscript. Specifically, we added concise background on transcription factor–associated fragmentation and repeat-element features to justify their inclusion and to improve the conceptual flow of the Introduction.

Secondly, we carefully reviewed and updated the references throughout the Introduction. The original References #4 and #5 were replaced with three more appropriate sources (Lines 131–135)—de Abreu et al. (The Oncologist, PMID: 40680231), Liu et al. (Ann Oncol, PMID: 32165213), and Jamshidi et al. (Cancer Cell, PMID: 36372334)—as these studies more accurately support the limitations of mutation-based assays (low VAF requiring deep sequencing) and methylation-based assays (modest Stage-I sensitivity and substantial computational burden).

Thirdly, we added a comprehensive comparison of breast cancer early-detection approaches in the revised manuscript. **Supplementary Table 6** now summarizes major diagnostic modalities—including mammography and ultrasound; mutation-based cfDNA assays; bisulfite-based, enzymatic, and ONT methylation assays; and leading fragmentomic platforms such as DELFI, CRAG, SPOT-MAS, cfFrag, GALYFRE, and TAPS—detailing their blood-volume requirements, workflow complexity, assay cost, and diagnostic performance. This table provides a breast-cancer-focused benchmark for evaluating the performance and practical advantages of TuFEst.

We believe our revisions substantially enhance the introductory framing and provide readers with a clearer understanding of the motivation, innovation, and clinical value of our work.

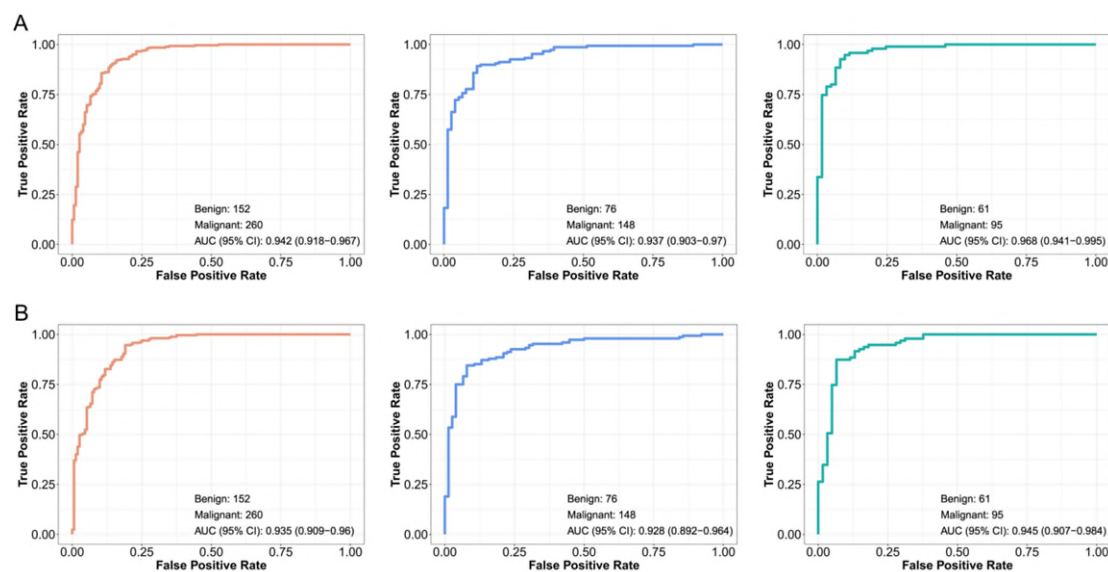
Point 3

At line 194, the authors describe filtering transcription factors (TFs) enriched in breast cancer using a breast cancer cell line. It would be helpful to include a discussion on how incorporating TFs enriched in benign conditions might influence model

performance.

Response: We thank the reviewer for raising this important point. The TFs analyzed in our study are specifically associated with breast cancer malignancy. Our selection criteria were intentionally designed to identify TFs linked to malignant biological processes, as the primary objective of the model is to distinguish malignant from benign states (PMID: 36399356). As a result, the malignant-associated and benign-associated TF groups are mutually exclusive by definition.

In addition, we conducted a dedicated ablation study comparing the full TuFEst model with a “Non-TF” model in which all TF-related coverage features were removed. The results, presented in **Rebuttal Figure 1**, demonstrate that incorporating TF features substantially enhances discriminative performance across the training, internal validation, and external validation cohorts.



Rebuttal Figure 1. Ablation analysis evaluating the contribution of TF-associated fragmentomic features to TuFEst classification performance.

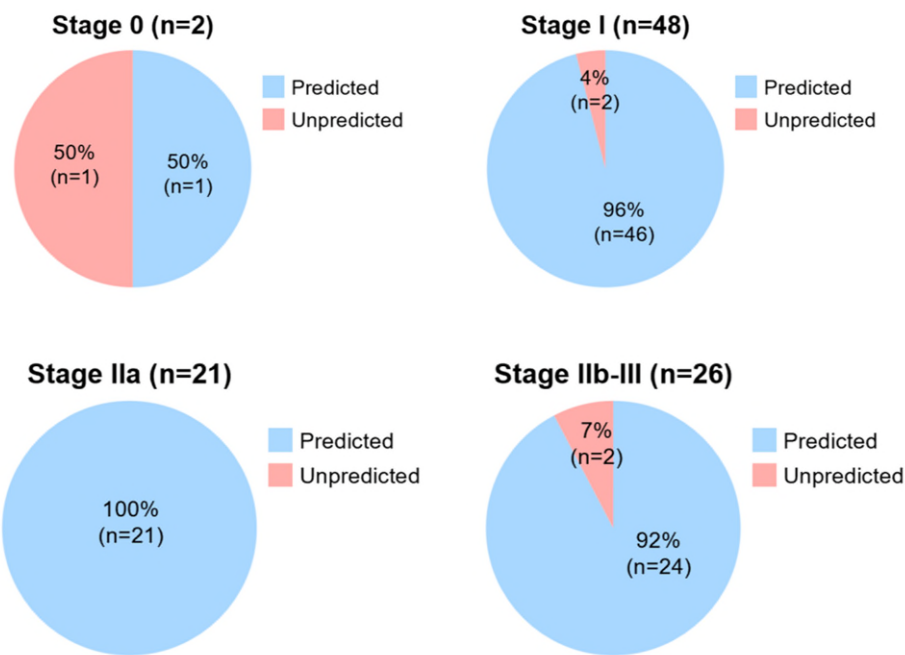
ROC curves evaluating the discriminative performance of SEM with (A) or without (B) TF-related coverage features across the training, internal validation, and external validation databases.

Point 4

In Figure 3K, the model correctly identified 25 out of 26 samples. Given the importance

of early-stage detection—both in this dataset and in the broader clinical context—it would be valuable to present the model’s performance specifically on early-stage samples, or stratified by stage. Also, while the 25/26 result in Figure 3K suggests outstanding early detection performance, a discussion comparing this result to the model’s performance on the broader dataset would provide important context and help assess the generalizability of the approach.

Response: We appreciate the reviewer’s emphasis on early-stage detection. We have performed an expanded stage-stratified analysis of TuFEst performance across Stage 0, I, IIa, and IIb–III samples in the external validation dataset. As shown in **Rebuttal Figure 2**, the model achieved consistently high detection accuracy across all early-stage categories: 96% in Stage I (46/48 correctly identified), 100% in Stage IIa (21/21). These stage-specific results place the 25/26 detection shown in Figure 3K into broader context and demonstrate that the strong performance observed in that subset is not an isolated finding but is consistent with robust detection across the full staging spectrum.

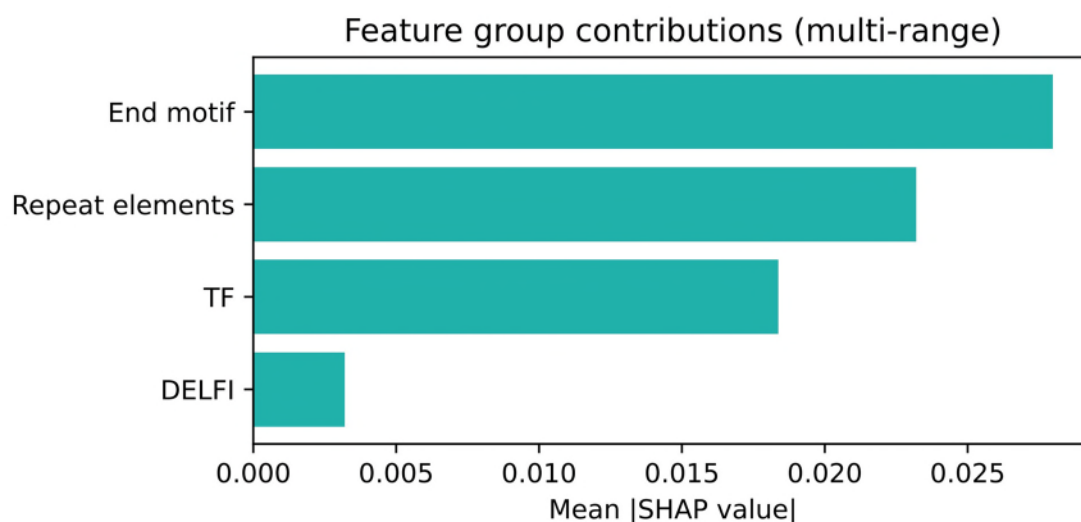


Rebuttal Figure 2. Stage-stratified performance of the TuFEst model in the external validation cohort.

Regarding Figure 3B, it would be helpful to present the individual performance of each modality (DELFI, end-motif, and repeat elements) at least in the Supplementary Materials. This would allow readers to better assess the relative significance of each modality and their respective contributions to the performance of the combined model.

Response: We thank the reviewer for this insightful suggestion. Clarifying the relative contributions of individual fragmentomic feature classes is indeed essential for both biological interpretability and future assay optimization. To address this point, we conducted additional analyses to systematically evaluate the predictive performance of each feature category in isolation. Specifically, we assessed models constructed using only DELFI fragmentation profiles, end-motif signatures, or repeat-element features. As shown in the newly added **Supplementary Figure 1**, each modality exhibited meaningful discriminatory power when evaluated on its own, and we now provide these results alongside the combined model shown in Figure 3B-D (Lines 264-268). Notably, the integrated TuFEst model consistently outperformed all single-modality models across cohorts.

Model interpretability analysis using SHapley Additive exPlanations (SHAP) further revealed that end-motif features contributed the largest mean absolute SHAP values, followed by repeat elements, TF signals, and DELFI profiles (**Rebuttal Figure 3**). These results indicate that all feature classes provide complementary and nonredundant biological information, and their integration is responsible for the superior performance of the combined model.



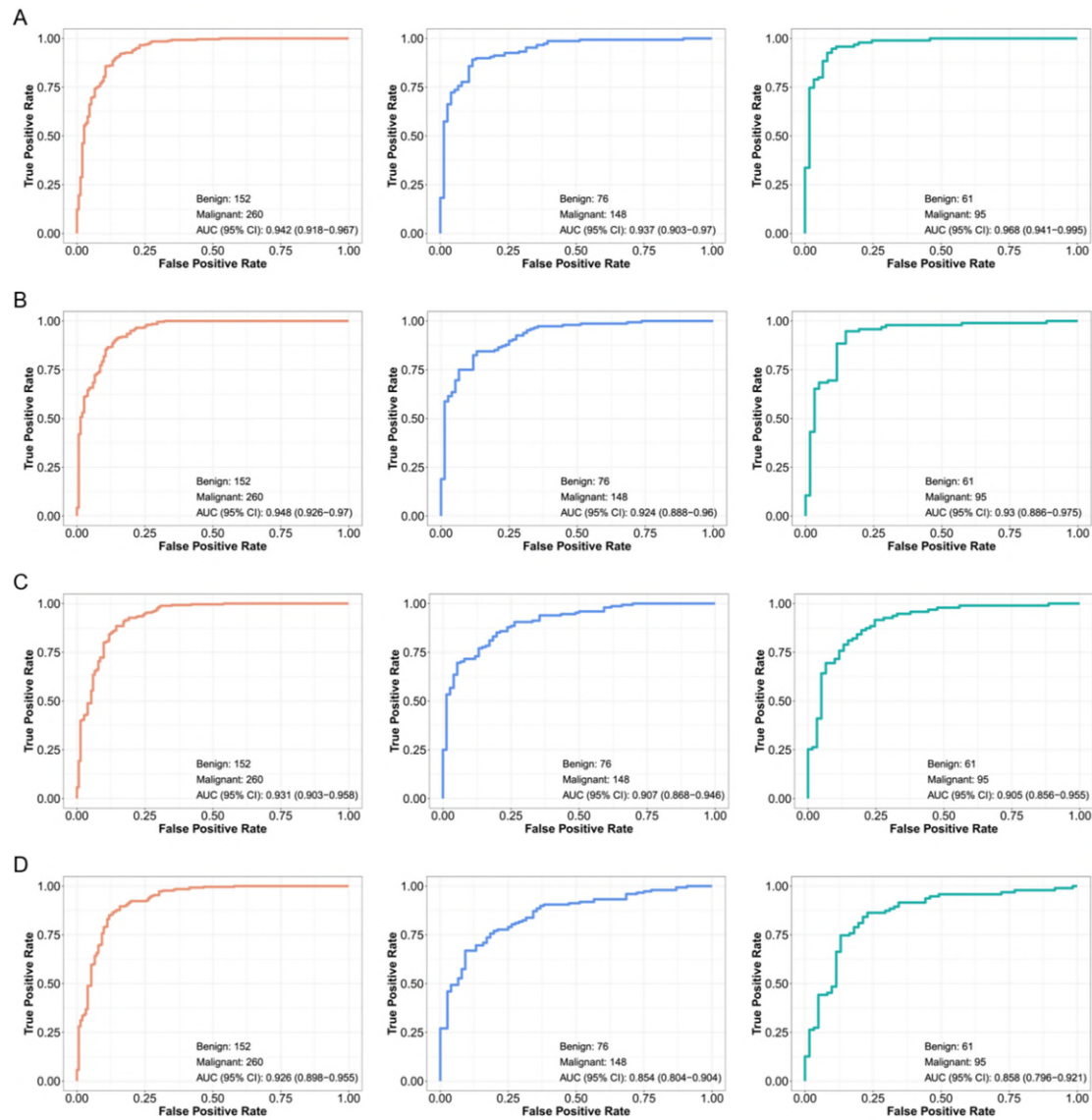
Rebuttal Figure 3. SHAP-based interpretability analysis of different fragmentomic features.

Point 6

The authors highlight the advantage of their approach as being suitable for ultra-low coverage settings. To support this claim, it would be helpful to present the coverage distribution across samples. As the authors note, lower coverage is indeed advantageous in clinical situations due to constraints in blood volume and cost. Therefore, it would strengthen the manuscript to include an analysis of breast cancer detection performance across different coverage levels or by down-sampling samples to low coverages (0.1~1X). If feasible, similar analyses for the other models presented in the manuscript would also be valuable. Additionally, a discussion explaining why low-pass sequencing seems to be required.

Response: We thank the reviewer for this valuable suggestion. We have now incorporated the coverage distribution for all samples and performed the requested down-sampling analysis from 2× to 1×, 0.5×, and 0.1×. As shown in **Rebuttal Table 1**, **Supplementary Table 3**, and **Rebuttal Figure 4**, TuFEst exhibited highly stable performance at 1× and 0.5×, with only minimal changes in AUC ($\Delta\text{AUC} \leq 0.02$) and sensitivity ($\leq 3\%$). A marked decline in performance was observed only at 0.1× coverage, indicating that sequencing depths in the range of 1–2× offer an optimal balance between assay accuracy and clinical feasibility. These results have been incorporated into the revised Results section (Lines 273–283).

We have also added a brief explanation in the Discussion regarding the rationale for low-pass sequencing in fragmentomics as follows (Lines 516–521): “Low-pass sequencing is inherently well suited for fragmentomics because its biological signal arises from genome-wide coverage variability, nucleosome-positioning patterns, and sequence-context features, rather than from high-depth variant detection such as single-nucleotide variants or variant allele frequencies. Importantly, increasing sequencing depth yields diminishing returns, as fragmentomic features tend to plateau once sufficient coverage uniformity has been achieved.”



Rebuttal Figure 4. Robustness of TuFEst performance across reduced sequencing depths.

ROC curves evaluating the diagnostic performance when reducing LP-WGS coverage from 2× (A) to 1× (B), 0.5× (C), and 0.1× (D) across the training, internal validation, and external validation databases.

Point 7

The authors claim that alternative technologies require high costs. To support this point, it would be helpful to include a comparison of the authors method to other existing methods regarding sequencing or assay costs.

Response: We thank the reviewer for this constructive suggestion. In the revised

manuscript, we have added a comprehensive comparison of sequencing and assay costs across major early breast cancer detection technologies, including mammography, ultrasound, mutation-based cfDNA assays, methylation-based assays (bisulfite, enzymatic, and nanopore-based), and fragmentomic approaches. These data are summarized in the newly added **Supplementary Table 6**, which outlines differences in required blood volume, workflow complexity, and estimated per-sample cost.

As highlighted in the Results and Discussion, TuFEst operates at the lowest sequencing depth (approximately 2×) and requires minimal sample input, resulting in substantially lower overall assay cost compared to other liquid biopsy methods. These advantages support the feasibility of our approach for large-scale population screening and implementation in resource-constrained clinical settings.

Point 8

Figure 2A appears to show less centralized plots across samples compared to DELFI. Even in the healthy group, the patterns seem less centralized compared to the DELFI healthy reference. It would be helpful for the authors to provide a rationale or explanation for this observation.

Response: We thank the reviewer for this thoughtful observation. To clarify this point, we performed an additional comparison and adjusted the visualization accordingly. The apparent “less centralized” pattern in Figure 2A arises primarily from differences in scaling and normalization between TuFEst and DELFI. In original Figure 2A, the visualization used z-score-standardized values, which expanded the dynamic range of signals across genomic bins and made sample-to-sample variation more visually pronounced—even among healthy controls. To address this, we re-generated the plots using raw coverage values instead of z-scores. As shown in **the revised Figure 2A**, this adjustment produces more centralized patterns without altering the underlying biological signal or affecting model performance.

Pont 9

At lines 187–189, the manuscript suddenly shifts to discussing transcription factors

(TFs), which may be confusing readers with the rationale for incorporating TF features. Providing additional background or a brief explanation would improve readers understanding. Similarly, at line 398, it is hard to get reasons why RNA-seq was chosen for downstream analysis, especially given the availability of other molecular data types such as methylation, SNVs, and CNVs. Although the rationale becomes somewhat clearer later in the Discussion (starting at line 478), the authors should consider introducing the reasoning earlier in the manuscript. A brief explanation of why xCell and ESTIMATE were selected—and how they support the study objectives—would also be helpful.

Response: We thank the reviewer for highlighting the need for clearer conceptual transitions. We have now revised the manuscript to provide explicit background and rationale at each of the indicated points.

First, we added a bridging explanation to clarify the rationale for incorporating transcription factor (TF)–associated features. Because cfDNA fragmentation patterns reflect nucleosome positioning and transcription factor occupancy, alterations in TF binding can leave characteristic footprints on plasma DNA. Prior fragmentomic studies have demonstrated that TF-site coverage is informative of regulatory activity and can enhance cancer detection performance (PMID: 31604930; 41043439). We now introduce this mechanistic rationale earlier in the manuscript to improve continuity when TF-related features are first presented (Lines 196–200 in the revised version).

Second, we clarified the reasoning behind selecting RNA-seq for downstream biological interpretation, despite the existence of alternative molecular data types such as methylation profiles, SNVs, and CNVs. Fragmentation patterns are closely linked to chromatin accessibility and transcriptional output; thus, RNA-seq provides a mechanistically grounded, orthogonal perspective for interpreting the biology underlying different TuFest Cancer Scores. We have now introduced this rationale where RNA-seq analyses first appear, rather than reserving it for the Discussion (Lines 426–429 in the revised version).

Finally, we added a brief explanation of why xCell and ESTIMATE were chosen for immune and stromal deconvolution (Lines 436–437 in the revised version). These

tools are widely used and well-validated for inferring tumor microenvironment composition from bulk RNA-seq, and they directly support our objective of connecting cfDNA fragmentation signals with underlying immune and stromal states.

Point 10

In the manuscript showed the detection of lymph node metastasis. It would be helpful for the authors to clarify whether lymph node metastasis could be detected within a unified model along with other types of metastasis (Figure 4). If this is not feasible, a discussion explaining the limitations or reasons for separating these models would strengthen the manuscript.

Response: We sincerely thank the reviewer for this thoughtful comment and appreciate the opportunity to clarify this point. We apologize for any ambiguity in the original presentation that may have contributed to misunderstanding. Figure 4 does not present a metastasis-prediction model. Instead, it reports the performance of TuFEst in predicting the molecular subtypes of metastatic lesions as defined by ER, PR, and HER2 status. Consequently, Figure 4 is unrelated to lymph-node or distant metastasis.

Regarding LN metastasis, our current analysis focuses specifically on identifying patients who are clinically node-negative (cN0) in early-stage breast cancer. The TuFEst-LN model demonstrates a high NPV, which is clinically meaningful because reliably excluding LN involvement may help reduce unnecessary invasive axillary procedures. The lower PPV, however, indicates that needle biopsy remains necessary when metastasis is suspected, consistent with standard practice.

The reviewer also raises an important point regarding whether LN metastasis and other metastatic sites could be predicted within a unified framework. At present, this is not feasible due to the limited number of patients with visceral or distant metastasis in our dataset, which constrains model development. Additionally, lymph-node metastasis and visceral/distant metastasis may generate distinct fragmentomic signatures due to differences in tumor burden and microenvironmental properties, further justifying separate modeling frameworks at the current stage. We agree that constructing an integrated metastasis-prediction model is an important and promising direction, and this

will be explored in future studies as larger and more comprehensive datasets become available.

Point 11

In the Discussion section (line 436), the authors mention the limitations of bisulfite-based methods. However, more recent approaches such as enzymatic C-to-T conversion and Oxford Nanopore Technologies (ONT)-based methylation profiling offer viable alternatives that may mitigate such limitations. The authors should provide more precise, performance-based evidence to support their claims. Furthermore, performance comparisons with existing studies—ideally in tabular format—would enhance the manuscript. In particular, at line 464, the authors compare their model to studies in other cancer types. It would be more clear to include direct comparisons with recent liquid biopsy studies of breast cancer.

Response: We thank the reviewer for this valuable suggestion. In the revised Discussion (Lines 508–516), we now provide a more precise, performance-based summary of recent alternatives to bisulfite conversion, including EM-seq, TAPS, and ONT-based direct methylation profiling. In addition, we have created a new **Supplementary Table 6** that systematically compares early breast-cancer detection performance and per-sample assay cost across major liquid biopsy platforms and imaging modalities. This table enables direct comparison between TuFest and recently published breast-cancer liquid biopsy studies, addressing the reviewer’s request for clearer benchmarking. We have also updated the Discussion to contextualize TuFest’s cost–performance characteristics relative to these existing methods, while taking care to emphasize that each modality has distinct advantages and trade-offs.

Point 12

At line 438, the authors refer to their approach as “AI-based.” However, given the methodological context, it may be more appropriate to describe it as “machine learning–based,” as the use of “AI” could be considered too advanced or potentially misleading in this case.

Response: We thank the reviewer for this important clarification. We agree that the term “AI-based” may overstate the methodological scope of our work and that “machine learning–based” more accurately reflects the approaches employed in this study. Accordingly, we have replaced “AI-based” with “machine learning–based” at the indicated location and have carefully reviewed the entire manuscript to ensure consistent and precise use of terminology throughout.

Point 13

At line 457, the authors mention using a tumor score, but what is the corresponding limit of detection in terms of variant allele frequency (VAF)? A brief discussion on this point would clarify the model’s sensitivity.

Response: We thank the reviewer for raising this important question regarding the model’s detection sensitivity. We agree that the limit of detection (LOD) in terms of variant allele frequency (VAF) is an essential parameter when interpreting the performance of sequencing-based assays.

Existing literature provides useful benchmarks for VAF LOD in targeted sequencing panels. For example, Petrackova et al. (PMID: 31552176), using a binomial probability model, recommend a minimum depth of $\geq 1,650\times$ and at least 30 mutant reads to reliably detect variants at $\geq 3\%$ VAF. Jennings et al. (PMID: 28341590; 33486075) similarly suggest coverage levels of 250–500 $\times$ for clinical-grade targeted NGS.

However, these standards apply specifically to deep targeted sequencing. The low-pass whole-genome sequencing (LP-WGS) approach used in our study (average coverage $\sim 2\times$) operates under a fundamentally different technical principle. At this coverage, the “tumor score” does not rely on detecting low-frequency single nucleotide variants (SNVs) or directly computing a VAF. Instead, it reflects tumor burden by capturing larger-scale genomic features—such as copy-number alterations (CNAs) and fragmentation-based chromatin signals—that can be robustly inferred even at ultra-low coverage.

For this reason, conventional VAF-based LOD benchmarks (e.g., 3% VAF

thresholds) are not applicable to LP-WGS fragmentomic analyses, and we have added a brief clarification in the revised manuscript accordingly (Lines 518–519).

Reviewer #2 (Remarks to the Author):

The manuscript introduces a new multi-feature cfDNA-based Breast cancer detection method. The method is trained and tested on a cohort of 503 cases and 289 controls and shows good classification performance. Additionally, they train a ML model that can perform molecular subtyping using the same features. The results are pretty good and of interest to people in the field, but the methods section is lacking details about the features and how the method was trained.

You use three different kinds of fragmentomics features. How many features do you end up with of each kind?

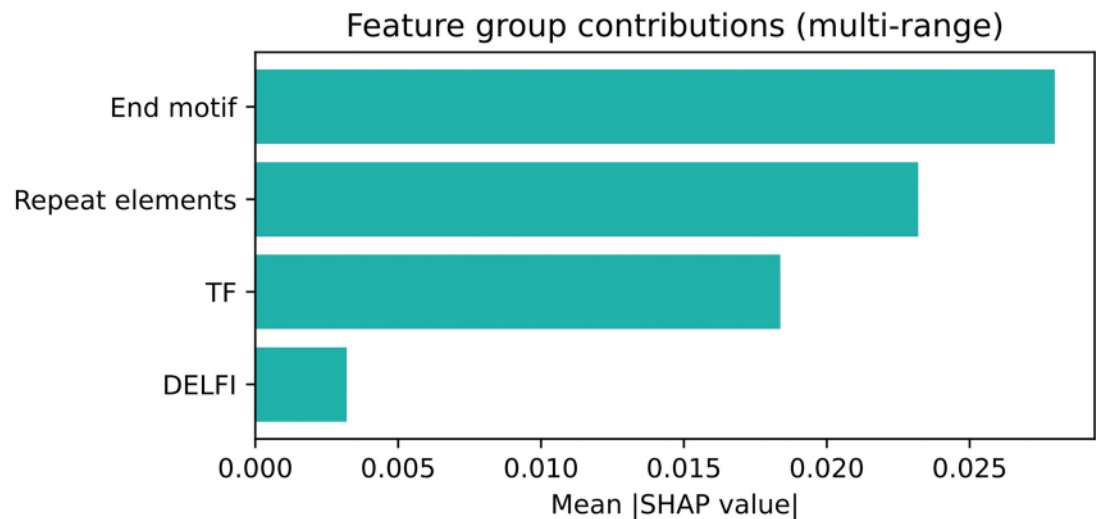
Response: Thank you for your comment. We ended up with the following number of features for each fragmentomics feature type, including TF (148 features) (PMID: 36399356), DELFI (515 features) (PMID: 31142840), End-motif (256 features for 4BP End-motif and 6 features for 6BP End-motif) (PMID: 32111602) and Repeat elements (1,279 features) (PMID: 38478628).

Have you tested which kind of fragmentomics features contribute the most to the performance of the method?

Response: We thank the reviewer for this insightful question. Clarifying the relative contributions of different fragmentomic feature classes is indeed critical for both biological interpretability and future assay optimization. In response, we conducted additional analyses to systematically evaluate the predictive performance of each modality independently. Specifically, we assessed models built using DELFI fragmentation profiles, end-motif signatures, and repeat-element features. As shown in the newly added **Supplementary Figure 1**, each modality exhibited detectable

discriminatory power on its own.

Model interpretability using SHapley Additive exPlanations (SHAP) further demonstrated that end-motif features contributed the largest mean absolute SHAP values, followed by repeat elements, TF features, and DELFI profiles (**Rebuttal Figure 3**). These results indicate that all feature classes provide complementary and nonredundant biological information, and their integration is responsible for the superior performance of the combined model.



Rebuttal Figure 3. SHAP-based interpretability analysis of different fragmentomic features.

Describing the motif analysis, you write that the 6-bp motifs are centered around the breakpoint. Is this also the case for the 4bp motifs, or are they only the end-motif?

Response: We thank the reviewer for this helpful clarification request. To clarify, the 4-bp motifs in our analysis are used exclusively as end-motifs (EDMs), representing the nucleotide sequences located precisely at the fragment termini. In contrast, the 6-bp motifs are defined as breakpoint-motifs (BPMs), which are centered around the predicted fragmentation breakpoints, as described in previous studies [PMID: 32111602; 35690859; 35780566]. Thus, the two motif classes reflect distinct biological signals: EDMs capture terminal sequence preferences at cleavage sites, whereas BPMs characterize the broader genomic context surrounding fragmentation breakpoints.

If you count motifs in both ends of the fragments, do you use the reverse complement of one of the ends so that the motifs are comparable?

Response: We thank the reviewer for raising this important methodological question. In standard end-motif analysis, reverse-complement transformation is often applied to one end to ensure direct comparability between the two fragment termini.

In our study, however, we intentionally analyzed the 5' and 3' end motifs separately, without applying reverse-complement merging. This design choice was motivated by the possibility that strand-specific cleavage preferences may exist, reflecting asymmetries in nuclease activity or DNA damage/repair processes. Treating the two ends independently preserves such potential biological signals, which would be lost if the ends were collapsed through reverse complementation.

Accordingly, our 256 end-motif (EDM) features represent the full set of 4-bp motifs derived independently from both fragment ends, providing a more comprehensive view of possible asymmetric fragmentation patterns, as also suggested by recent fragmentomic studies (PMID: 32111602; 35690859; 35780566).

There are no details about the different ML methods that are trained provided in the manuscript. I looked at the linked GitHub repo, but as far as I can see, that repo only contains code for plotting. You should both add more details about the parameters and implementations of the methods to the manuscript and add the code to GitHub to make the details regarding the training and testing of the models available.

Response: We sincerely thank the reviewer for this valuable comment and fully agree that improving methodological transparency is essential. In the revised manuscript, we have expanded the Methods section (Lines 678–719 in the revised manuscript) to provide detailed descriptions of the machine-learning workflow. Specifically, we now clarify that: “A SEM integrated predictions from five base learners-SVM, GLM, GBM, XGBoost, and MLP-using a logistic regression meta-learner. The pipeline included feature standardization and was evaluated through 10-fold stratified cross-validation. Performance metrics were calculated for each fold, and probability thresholds were established at specificity levels of 0.9 and 0.95. The ensemble demonstrated strong

discriminative ability (AUC = 0.943), with all predictions and sample information systematically recorded. The complete pipeline was saved to support potential clinical implementation.”

To further ensure full transparency and reproducibility, we have uploaded the complete machine-learning codebase—including training, evaluation, and prediction scripts—to a dedicated GitHub repository:

https://github.com/yanwenfei0926/BC_CFDNA/tree/main/BC_Model/

This repository now contains all scripts necessary to reproduce the results presented in the manuscript, addressing the reviewer’s concerns.

For instance, is it a linear SVM you test, or do you use a kernel? Do you use cross-validation to fit any meta-parameters for any of the ML methods, or do you use default parameters? You end up using a Stacked Ensemble (I would write out the abbreviation the first time you use it), but that is just a method for combining other ML models, and you do not say anything about which models you are combining (is it the 5 other mentioned models?).

Response: We sincerely thank the reviewer for these important clarifying questions. We have now revised the Methods section to explicitly provide the requested technical details.

First, regarding the SVM implementation: we used a linear SVM, consistent with prior work demonstrating improved stability in high-dimensional settings and to reduce the risk of overfitting with limited samples. We now explicitly report this choice in the revised manuscript (Lines 685–692).

Second, for all machine-learning methods—including SVM, GLM, XGBoost, MLP, and GBM—we employed a nested cross-validation strategy to tune hyperparameters. Specifically, a 10-fold stratified outer loop was used for performance evaluation, while an inner loop performed grid-search-based hyperparameter optimization on each training fold. This procedure reduces optimistic bias and ensures that the reported metrics reflect generalizable performance. All hyperparameter grids and training scripts are available in the GitHub repository:

https://github.com/yanwenfei0926/BC_CFDNA/tree/main/BC_Model/

Third, we appreciate the reviewer's suggestion to clarify the composition of the Stacked Ensemble Model (SEM) and to write out the abbreviation at first use. As now stated in the revised text, the SEM combines predictions from the five base learners evaluated in this study (GLM, linear SVM, XGBoost, MLP, and GBM). A logistic regression meta-learner was then trained on the out-of-fold predictions from these base models to generate the final ensemble prediction. We have updated the manuscript to clearly describe this structure (Lines 713–719).

We thank the reviewer again for these insightful questions, which have helped us substantially improve the clarity and transparency of our methodological description.

The method is called TuFEst (Tumor Fraction Estimator), but the manuscript only reports on classification performance; there are no plots showing an estimated tumor fraction compared to some independent estimate of the tumor fraction in the cfDNA samples.

Response: We thank the reviewer for this insightful observation. We fully acknowledge that the name “Tumor Fraction Estimator (TuFEst)” may suggest direct quantification of tumor fraction. In the current study, however, TuFEst is used primarily as a classification model, and its output—referred to as the cancer score (Figure 3H–J)—represents a probability value (cutoff = 0.5) used for malignancy prediction rather than an explicit estimate of the percentage of tumor-derived cfDNA.

At present, we do not have an independent ground-truth measure of tumor fraction for these clinical samples (e.g., through deep targeted sequencing or orthogonal cfDNA assays), and therefore cannot provide direct tumor fraction-to-tumor score comparisons. For this reason, we have focused on reporting classification performance.

We retained the name “TuFEst” because the cancer score reflects a surrogate measure of tumor-derived DNA burden, derived from fragmentomic features known to correlate with tumor fraction in prior studies. This framework provides the conceptual basis for extending TuFEst into a true quantitative tumor fraction estimator in future work. We have revised the manuscript to clarify this distinction and avoid ambiguity

(Lines 672–677 in the revised version).

Reviewer #2 (Remarks on code availability):

As far as I can see, that repo only contains code for plotting. I could not find code for training the different models mentioned in the manuscript.

Response: Thank you for carefully examining our code repository and for raising this important point. We apologize for the earlier omission. We have now uploaded the complete code for training all models described in the manuscript to the following directory in the GitHub repository:

https://github.com/yanwenfei0926/BC_CFDNA/tree/main/BC_Model/

Reviewer #4 (Remarks to the Author):

This is a well written manuscript demonstrating the power of cfDNA fragments to detect breast cancer as well as determine subtype and lymph node status.

It was a well conducted study using both benign and malignant breast tissue.

I think that this study will be of interest to researchers and clinicians involved in early breast cancer detections and treatment.

I have very minor concerns:

1. I could not locate Table 1 cited on line 209.

Response: First, we sincerely thank the reviewer for the positive and encouraging comments. We truly appreciate the reviewer's recognition of our study design. In terms of this issue, we apologize for the oversight regarding Table 1, which has now been moved to the correct location (Line 1093). We have carefully checked all table and figure placements to ensure consistency between the text and the manuscript layout.

2. there is no discussion of tumors which are triple positive, whether they were included or excluded in the analyses.

Response: We thank the reviewer for this helpful comment. In our study, molecular subtyping was based on ER, PR, and HER2 expression status, following standard therapeutic categories (HR⁺/HER2⁻, HER2⁺, and TNBC). Under these commonly used clinical definitions, triple-positive breast cancers (TPBC; ER⁺/PR⁺/HER2⁺) are not treated as a separate subtype but are included within the HER2⁺ group, as HER2-targeted therapy primarily determines treatment stratification.

In our cohort, the number of TPBC cases was relatively small, which further limited the statistical power to train and evaluate a dedicated TPBC class. For these reasons, TPBC tumors were grouped within the HER2⁺ subtype during model development. We have now explicitly clarified this in the revised manuscript.

We also acknowledge this as a limitation (Lines 576–578 in the revised version) and note that future studies with larger cohorts will allow us to assess whether cfDNA fragmentomic signatures can further distinguish TPBC as a biologically or clinically distinct subgroup. We appreciate the reviewer's suggestion, which has helped improve the clarity of our study design and highlighted an important direction for future research.

Rebuttal Table 1. Summary of sequencing depth for the BC samples

Sample	Group	Sequencing depth (2×)	Sequencing depth (1×)	Sequencing depth (0.5×)	Sequencing depth (0.1×)
BC0037	TD	1.694242881	0.951577558	0.4850384	0.108107956
BC0040	TD	1.58181094	0.94171843	0.489342553	0.094039362
BC0067	TD	1.895820128	0.962214531	0.519774156	0.094017244
BC0094	TD	1.606148441	0.941788716	0.513519771	0.103225262
BC0155	TD	2.038081888	0.97695245	0.504218276	0.091545435
BC0185	TD	1.663852768	0.943486624	0.513121055	0.105527238
BC0197	TD	1.711021083	1.013652873	0.488935184	0.095133166
BC0201	TD	1.59800978	0.974306793	0.519856471	0.096442327
BC0228	TD	1.713700039	1.062876519	0.487917778	0.108023545
BC0369	TD	1.747760615	0.985011208	0.481834459	0.099939588
BC0428	TD	1.833142622	1.0692112	0.483291355	0.101944166
BC0477	TD	1.807636224	1.044949854	0.511632186	0.103256847
BC0486	TD	1.634941244	1.062421713	0.50696022	0.108071243
BC0514	TD	1.789858026	1.03485402	0.510261083	0.096700115
BC0526	TD	2.218979038	1.048482623	0.512964886	0.093646562
BC0648	TD	2.066962097	1.03528404	0.488311758	0.109426693
BC0652	TD	1.571961164	1.042614704	0.490718965	0.106895181
BC0656	TD	1.787282179	0.982796559	0.497320632	0.103192965
BC0722	TD	2.001073831	0.937402652	0.481330104	0.094701937
BC0088	TD	1.947287235	1.062377495	0.515883808	0.103072603
BC0243	TD	1.526094925	1.016213646	0.518175903	0.102127617
BC0257	TD	2.014403263	1.050048454	0.501626429	0.090364408
BC0422	TD	2.111057951	1.067216855	0.492942556	0.104370444
BC0664	TD	1.776312556	0.989232305	0.514940467	0.100464739
BC0098	TD	2.176362122	1.01608481	0.498116447	0.101793413
BC0150	TD	1.875862247	0.975225681	0.500680823	0.095034715
BC0183	TD	1.500098388	1.008877106	0.508721051	0.109682554
BC0281	TD	2.027976363	1.019691708	0.504271639	0.109157732
BC0316	TD	1.446061785	1.021500454	0.494057286	0.102198684
BC0358	TD	1.881498158	0.988284066	0.495403175	0.094908923
BC0419	TD	1.858163622	1.065541387	0.499871384	0.097529615

BC0423	TD	2.01126981	1.009877224	0.495125101	0.102697745
BC0431	TD	2.172913224	0.961331357	0.500120693	0.107593621
BC0472	TD	2.098908664	0.938488386	0.483593043	0.10453493
BC0490	TD	1.424777217	1.0337179	0.516501354	0.095559435
BC0509	TD	1.885769096	1.020450358	0.50120931	0.105815373
BC0511	TD	1.673688951	0.932876699	0.508689812	0.107694092
BC0524	TD	1.751564496	0.996488249	0.481232006	0.105534155
BC0585	TD	1.621676549	1.022706807	0.501076291	0.10245335
BC0670	TD	1.666067643	1.043127934	0.488017836	0.092150101
BC0835	TD	1.638923132	1.036858925	0.50285052	0.10904699
BC0840	TD	1.696824383	0.961120382	0.509192421	0.095280119
BC0001	TD	1.866062268	1.001039978	0.507125573	0.107756391
BC0117	TD	2.002676775	1.051501277	0.493680086	0.10155195
BC0149	TD	1.912907051	1.020275139	0.498214472	0.095119158
BC0173	TD	1.644078384	0.972047037	0.504275932	0.097334115
BC0311	TD	1.563047803	0.999135541	0.492269525	0.109749603
BC0531	TD	1.791599762	0.964516654	0.496370453	0.095246396
BC0682	TD	2.244345019	1.031891268	0.490547531	0.090005469
BC0709	TD	2.459077136	0.940004402	0.503213595	0.09347113
BC0714	TD	2.157216432	1.042158247	0.483025341	0.102034571
BC0725	TD	1.905975345	0.945607121	0.519631306	0.090952828
BC0066	TD	2.009499593	1.056983105	0.4839697	0.106803861
BC0129	TD	2.109968085	1.015955702	0.515043296	0.094782807
BC0237	TD	1.609180398	0.994600494	0.508596221	0.103544184
BC0245	TD	1.863615011	1.020033512	0.504746618	0.106654029
BC0246	TD	1.879691648	0.93388571	0.486340868	0.105456645
BC0248	TD	2.024807773	0.973505184	0.519780788	0.097396401
BC0269	TD	1.672359331	0.932629142	0.515423964	0.098363579
BC0403	TD	1.934649858	1.027311965	0.490985682	0.101143614
BC0430	TD	1.9695032	1.021337014	0.515144626	0.097424281
BC0542	TD	1.966123477	1.047100102	0.485388048	0.092360527
BC0665	TD	1.759217174	0.990728278	0.489497391	0.108885256
BC0690	TD	2.594670422	1.020051517	0.48513538	0.104326285

BC0694	TD	2.258347121	0.987879021	0.485714782	0.097623706
BC0135	TD	1.692432834	0.946378491	0.492525445	0.103329678
BC0159	TD	1.944846963	1.035754698	0.503436344	0.104215364
BC0162	TD	1.897695546	1.033131849	0.504451762	0.095704276
BC0171	TD	1.923583559	1.054545082	0.517769569	0.105251767
BC0219	TD	1.726032774	1.049255085	0.48734664	0.093600294
BC0266	TD	2.061564859	1.04035925	0.492110171	0.09446752
BC0276	TD	1.825986282	0.961418593	0.487688593	0.104664733
BC0287	TD	1.732294018	1.029058576	0.482933778	0.093148077
BC0293	TD	1.879645843	0.95218494	0.482988078	0.096493669
BC0302	TD	1.518967705	1.018550582	0.490857982	0.101563931
BC0333	TD	1.663188319	0.975398087	0.519340125	0.090640964
BC0353	TD	1.415660262	0.997624761	0.51961484	0.098283582
BC0455	TD	1.677493251	1.037975648	0.503895919	0.105863394
BC0460	TD	2.252376184	1.01657298	0.506257438	0.097912556
BC0487	TD	1.89425836	1.056202688	0.507506141	0.092225616
BC0506	TD	1.689302073	0.956834174	0.516579636	0.095376734
BC0535	TD	1.670548223	1.020249263	0.492824637	0.109898492
BC0669	TD	1.866669347	1.06630177	0.512674635	0.091899686
BC0703	TD	2.130346012	1.058393911	0.507450563	0.102455881
BC0051	TD	3.623029724	1.066887708	0.491896325	0.105029769
BC0126	TD	1.643954231	1.04689304	0.504134737	0.106256337
BC0210	TD	1.758234585	1.060890287	0.515073615	0.099850589
BC0225	TD	1.741549598	0.982612876	0.511486071	0.095722187
BC0230	TD	1.933148742	1.023421977	0.496127385	0.09839319
BC0292	TD	1.604605311	1.059507884	0.505154916	0.096752354
BC0314	TD	1.689334586	1.020836493	0.519617409	0.092668581
BC0329	TD	1.287066542	0.994412173	0.488158185	0.109221633
BC0368	TD	1.811615465	1.01163283	0.513558925	0.094974584
BC0389	TD	2.016577615	1.044280515	0.488241731	0.105027795
BC0503	TD	1.625733207	1.001200028	0.503819568	0.105799044
BC0654	TD	1.5957846	0.989778002	0.484735867	0.109803023
BC0683	TD	2.21496782	0.964210089	0.484011342	0.102469724

BC0684	TD	2.39069185	1.046719684	0.498222747	0.091536265
BC0711	TD	1.689229029	1.068029367	0.503488193	0.094762752
BC0716	TD	1.858393461	1.010777894	0.501493278	0.10576547
BC0721	TD	2.085614059	1.067927696	0.516058191	0.098722264
BC0093	TD	1.578350742	0.973017188	0.508451984	0.102695007
BC0137	TD	1.865831271	0.988809556	0.480720551	0.102951983
BC0420	TD	1.240752795	1.031645279	0.492699586	0.105952718
BC0533	TD	1.732379768	0.930906601	0.484467805	0.108793964
BC0659	TD	1.73717425	1.042962592	0.505379722	0.100504262
BC0108	TD	1.839683055	1.06602917	0.492509592	0.101049824
BC0139	TD	1.768949739	1.03063465	0.48434564	0.102471518
BC0238	TD	2.520283056	1.068270286	0.510009425	0.105133845
BC0263	TD	1.697176613	0.95768057	0.515450407	0.091830161
BC0265	TD	1.974835814	0.975903158	0.510922232	0.099650519
BC0318	TD	1.705350556	1.012555928	0.505342428	0.100237474
BC0425	TD	1.699648344	0.97120052	0.518086829	0.106349641
BC0437	TD	2.089186299	1.041410487	0.497855146	0.093652155
BC0447	TD	1.991542054	1.066525662	0.491011962	0.108627778
BC0453	TD	1.999691742	1.062669034	0.481870887	0.10890857
BC0482	TD	1.367750773	0.949067431	0.510995327	0.108995238
BC0491	TD	1.930317556	0.944178555	0.510920225	0.091783346
BC0517	TD	1.913120587	1.059286971	0.495727281	0.094613386
BC0573	TD	2.301171572	1.007656515	0.494866912	0.102508064
BC0592	TD	1.804740311	1.062595938	0.508887262	0.104676599
BC0673	TD	1.796313044	0.930581566	0.49156034	0.109559297
BC0675	TD	2.031196726	0.939025428	0.504539622	0.107606194
BC0698	TD	2.720449906	1.054224863	0.485603156	0.109817857
BC0836	TD	1.92585049	1.029254133	0.488270204	0.105729976
BC0165	TD	1.430149833	1.049022418	0.512834433	0.102936089
BC0177	TD	1.735521809	1.039039741	0.502019487	0.092824628
BC0189	TD	1.899680058	0.935955135	0.500053639	0.106719864
BC0191	TD	1.891549257	0.943520706	0.514236134	0.093122527
BC0213	TD	1.631448698	1.029428891	0.500081438	0.109260336

BC0218	TD	1.672781649	1.001647836	0.496197538	0.102881387
BC0306	TD	1.595165322	1.03387194	0.50272605	0.103077047
BC0359	TD	1.699414689	0.935802272	0.501510265	0.100638247
BC0387	TD	1.445011398	0.947762777	0.485380915	0.101498999
BC0406	TD	1.44830085	1.020930635	0.482977617	0.090198382
BC0496	TD	1.867751575	0.991727197	0.500459809	0.091381849
BC0512	TD	1.868927951	1.023362603	0.499860513	0.101258292
BC0516	TD	2.157190533	1.058256078	0.495270128	0.102228658
BC0710	TD	2.45998395	1.065633063	0.492556216	0.096784581
BC0718	TD	2.185286556	0.971953279	0.495265926	0.106821881
BC0723	TD	1.940342024	1.060000483	0.482797133	0.106284173
BC0198	TD	1.925482163	1.031816586	0.51762058	0.104614739
BC0247	TD	1.895346953	0.948579956	0.482597077	0.108694442
BC0390	TD	2.007773737	1.039715375	0.49787303	0.09012463
BC0415	TD	1.870088577	0.960130982	0.504239049	0.095179924
BC0532	TD	1.825604383	0.998429285	0.506426888	0.097478651
BC0661	TD	1.740517708	0.954238541	0.482820086	0.105472763
BC0697	TD	2.751530915	0.972838056	0.490079399	0.099683108
BC0119	TD	1.56931334	1.021475385	0.511286696	0.100860831
BC0133	TD	1.896743156	1.032523842	0.48747874	0.103280408
BC0286	TD	1.385972656	1.057809469	0.480309127	0.105940617
BC0294	TD	1.976307759	0.984538432	0.514959023	0.10185913
BC0341	TD	1.36001393	0.956511436	0.487240981	0.09313497
BC0374	TD	2.166613744	1.036387973	0.490996134	0.096725246
BC0380	TD	2.126338389	1.022128433	0.506420931	0.100372655
BC0388	TD	1.954041454	0.98235498	0.51171128	0.093909737
BC0410	TD	3.297961395	1.03599692	0.507040846	0.099924334
BC0489	TD	2.082805088	0.94304939	0.498224378	0.10719341
BC0545	TD	1.543685545	1.033834313	0.505355832	0.102304937
BC0553	TD	1.958745695	0.932551434	0.491075688	0.107638111
BC0559	TD	1.848916728	1.017094441	0.500585745	0.095959412
BC0594	TD	1.865396651	1.045260116	0.502836773	0.091036499
BC0676	TD	1.755836868	0.972543399	0.512892736	0.09633266

BC0831	TD	1.924877872	1.044233836	0.517291524	0.099688462
BC0838	TD	1.885387736	1.024192251	0.489330832	0.108346097
BC0842	TD	1.895200825	0.950854011	0.506918016	0.106795227
BC0077	TD	1.800416783	0.971596266	0.491244453	0.109887591
BC0082	TD	1.817232256	0.944815817	0.513976006	0.10308059
BC0092	TD	1.979844838	0.953593002	0.503351815	0.105878521
BC0182	TD	1.791712023	0.97112508	0.50128505	0.098288272
BC0187	TD	1.931348087	1.039616433	0.485895113	0.094396635
BC0205	TD	1.673986001	0.956296781	0.508899664	0.09695768
BC0273	TD	2.120365712	0.953309577	0.515410658	0.09117401
BC0335	TD	1.764791533	1.020656664	0.519497715	0.090381129
BC0337	TD	1.590961182	1.002900182	0.507579673	0.105338713
BC0338	TD	1.885621217	1.019886654	0.493944756	0.093153348
BC0345	TD	1.70302109	1.003391958	0.501708033	0.097848429
BC0354	TD	1.544061364	1.001561736	0.508297804	0.1016413
BC0395	TD	1.190712368	1.042887907	0.500313799	0.102351102
BC0488	TD	1.832618341	1.009529983	0.50504728	0.092547012
BC0681	TD	1.656274301	0.973728103	0.505387915	0.109644595
BC0686	TD	2.775133063	0.950231639	0.518132154	0.100208039
BC0687	TD	2.582213889	1.0497797	0.518997724	0.100619772
BC0717	TD	1.690827802	1.059457509	0.496199233	0.108710149
BC0232	TD	2.063741977	0.9320341	0.486692385	0.107957772
BC0249	TD	1.827609541	1.066209443	0.513251146	0.104918637
BC0251	TD	2.116519475	1.048605534	0.48578629	0.094825355
BC0252	TD	1.815461067	1.059962326	0.510126655	0.106761564
BC0398	TD	1.607258695	0.93969278	0.515840816	0.09951952
BC0405	TD	1.845340578	1.059451793	0.48802816	0.105740018
BC0688	TD	2.430166808	1.010406352	0.519523778	0.100228155
BC0689	TD	2.505429221	0.997544689	0.50713301	0.102814663
BC0186	TD	1.87401714	0.990983544	0.488908719	0.107085705
BC0253	TD	1.83323318	1.028184956	0.49647724	0.107065229
BC0277	TD	1.968796609	1.043917284	0.48217737	0.107891538
BC0317	TD	1.894129551	1.064723339	0.495446791	0.097855825

BC0367	TD	2.277527414	1.06617506	0.503561909	0.107873065
BC0444	TD	1.424879819	0.948054049	0.511386831	0.093864205
BC0454	TD	1.826790516	0.957036854	0.511371102	0.104932455
BC0475	TD	1.854846497	0.935903278	0.507812628	0.107027153
BC0505	TD	1.874430669	1.027184389	0.503302552	0.093353309
BC0554	TD	1.695202227	1.02657053	0.514994785	0.094771517
BC0579	TD	2.021337243	0.944488834	0.512691236	0.096834855
BC0593	TD	1.761761313	1.015867999	0.489069912	0.091204409
BC0672	TD	2.007950961	1.031217516	0.518814492	0.104122192
BC0700	TD	2.585393125	1.031801109	0.506173038	0.105709699
BC0828	TD	1.765546895	0.99449362	0.518285326	0.105704813
BC0034	TD	1.930492397	0.936439591	0.483155204	0.094113348
BC0083	TD	1.679849105	1.021204246	0.501956255	0.100093967
BC0153	TD	2.09181548	1.033407475	0.495204901	0.106187603
BC0169	TD	2.004022737	1.066609713	0.494352424	0.107679533
BC0193	TD	2.075797204	0.958274266	0.488428519	0.102232059
BC0207	TD	1.336339093	1.048595563	0.493056715	0.095947665
BC0327	TD	2.117096154	1.032619557	0.501416919	0.107533027
BC0349	TD	1.995311352	0.980729533	0.516833694	0.109580184
BC0360	TD	1.687313394	0.959843441	0.511462354	0.105610075
BC0378	TD	1.889921526	0.960418661	0.515428505	0.105525174
BC0457	TD	1.709422279	1.042970717	0.487734073	0.099712385
BC0481	TD	2.071561167	1.033310598	0.508104973	0.090568851
BC0483	TD	1.609255049	0.988721546	0.509187571	0.094118631
BC0519	TD	1.866854637	1.020677257	0.493492372	0.091777841
BC0657	TD	1.759280126	0.948760772	0.49109665	0.097719799
BC0679	TD	2.437385592	1.027315389	0.516437654	0.108151011
BC0087	TD	3.355044407	0.975944388	0.508369032	0.101431219
BC0102	TD	1.644974993	1.003403055	0.515971646	0.10828597
BC0122	TD	2.141379647	0.999875454	0.485358111	0.106419467
BC0190	TD	2.018772945	1.049013968	0.511019468	0.106206175
BC0256	TD	1.782095299	0.962883754	0.516188638	0.10478425
BC0262	TD	1.701436512	1.069434288	0.519389822	0.098976154

BC0534	TD	2.174996757	0.965728794	0.504204269	0.103889029
BC0662	TD	1.745163736	0.957016237	0.500303778	0.108070905
BC0146	TD	1.701545788	1.010863453	0.514327655	0.099696822
BC0174	TD	1.759105689	1.067868532	0.501111865	0.090844001
BC0220	TD	2.021651627	0.999932139	0.504493811	0.104735186
BC0236	TD	2.035691192	1.064430538	0.489628886	0.109482344
BC0250	TD	2.051910643	1.003394778	0.502843916	0.108074259
BC0279	TD	1.743640932	1.059928363	0.518054384	0.09121331
BC0304	TD	1.663288767	1.014292618	0.500829177	0.107278956
BC0320	TD	1.92942777	0.965818834	0.496004162	0.107544539
BC0429	TD	1.78894591	1.004960612	0.510284524	0.104048396
BC0440	TD	1.886639782	0.93970981	0.490743097	0.104116427
BC0522	TD	1.753180407	0.960647221	0.505338359	0.104397953
BC0529	TD	1.859681078	0.972088794	0.515444093	0.109362107
BC0537	TD	2.038608706	1.020867888	0.482591769	0.095681537
BC0677	TD	1.743397068	1.063672453	0.514851752	0.10640411
BC0701	TD	2.427867806	1.045024862	0.480366302	0.107182479
BC0707	TD	2.059240992	1.067410907	0.495426961	0.108872437
BC0844	TD	1.969812102	1.050234008	0.51409361	0.099445124
BC0007	TD	1.88526843	1.052907145	0.515844349	0.093403321
BC0027	TD	1.589264626	0.945463606	0.514798927	0.099431043
BC0039	TD	1.996382207	0.948947022	0.509285406	0.101304435
BC0113	TD	2.171229187	1.020347921	0.510433708	0.109917528
BC0199	TD	1.788713516	0.9794249	0.490213604	0.09265974
BC0202	TD	1.983982587	1.025013993	0.502711765	0.101548212
BC0211	TD	2.058224299	1.051328997	0.486223371	0.109488497
BC0216	TD	1.670380418	0.993641855	0.48359698	0.105445606
BC0343	TD	1.723264573	1.018792075	0.505165524	0.10782149
BC0352	TD	2.183739445	0.991490805	0.489407452	0.107451172
BC0521	TD	1.878870122	1.059665905	0.487030409	0.107930696
BC0651	TD	2.036442986	1.032177082	0.489395619	0.100415414
BC0712	TD	2.206566194	0.985503829	0.513020381	0.103620986
BC0715	TD	2.19407262	0.973190007	0.499830163	0.093814445

BC0719	TD	1.563149375	0.950933537	0.489080975	0.106938692
BC0724	TD	1.933939758	1.035766027	0.488455553	0.105908262
BC0074	TD	1.674534185	0.945210181	0.493445847	0.107634041
BC0107	TD	2.031773969	1.057442869	0.507019486	0.094030639
BC0125	TD	1.74936193	1.008312036	0.512803303	0.099310628
BC0393	TD	1.371861651	1.053848073	0.497151175	0.100187039
BC0411	TD	1.708740026	1.025279185	0.502075538	0.109258774
BC0666	TD	1.543142357	0.996779685	0.48389037	0.106593852
BC0691	TD	2.46473891	0.981183252	0.519239769	0.100978739
BC0693	TD	1.920479934	1.061348028	0.515682702	0.099174707
BC0079	TD	1.746993041	1.048217057	0.490442293	0.099425229
BC0167	TD	3.314126035	0.972265813	0.483505653	0.094161886
BC0271	TD	1.772020352	1.038396863	0.503684411	0.105339948
BC0274	TD	1.848342281	1.039264956	0.517619903	0.105123302
BC0299	TD	1.730387993	1.052754874	0.496227579	0.100550582
BC0305	TD	1.830966205	0.937139561	0.501273622	0.108016076
BC0336	TD	1.73709998	1.019965943	0.491501056	0.109259516
BC0361	TD	2.508624518	1.02964283	0.494334658	0.107522716
BC0436	TD	1.659974951	0.98622607	0.486267142	0.107508439
BC0468	TD	1.95958864	0.997705386	0.491431014	0.096109784
BC0480	TD	2.104902562	0.983812908	0.486797694	0.091435086
BC0495	TD	1.956595721	1.048487685	0.514044671	0.092830526
BC0507	TD	1.684120603	1.064841452	0.485417603	0.105495515
BC0674	TD	1.785684898	0.982106578	0.486256903	0.102834056
BC0704	TD	2.576998209	0.994949175	0.510142073	0.097808572
BC0826	TD	1.683704322	0.995682225	0.500899006	0.101133075
BC0834	TD	1.472118924	0.94639453	0.519613026	0.099877408
BC0012	TD	1.821683516	0.959986688	0.503446175	0.103246113
BC0054	TD	2.087899343	0.958242479	0.482479605	0.108055294
BC0059	TD	1.971636377	0.935441753	0.487465986	0.108292693
BC0064	TD	1.986117504	0.956773069	0.490099405	0.097879438
BC0097	TD	1.765121876	0.98151763	0.507555459	0.098368377
BC0175	TD	1.900138968	1.069145845	0.512174651	0.107928944

BC0194	TD	2.158410406	0.997494044	0.504359177	0.106104913
BC0214	TD	1.942712881	0.974819092	0.481473899	0.098750629
BC0221	TD	2.347182639	1.005166639	0.504507524	0.096332736
BC0309	TD	1.567022101	1.048059558	0.514701514	0.09635691
BC0344	TD	2.137404952	1.064729972	0.484783574	0.091943153
BC0400	TD	1.754590887	1.036135529	0.509784268	0.108721348
BC0479	TD	1.763314634	0.954559801	0.49649633	0.094656187
BC0649	TD	1.626436473	1.047839523	0.485414159	0.105519283
BC0727	TD	1.865599202	1.030174458	0.493488889	0.093081901
BC0069	TD	1.305875994	0.941669062	0.486366315	0.10363702
BC0078	TD	2.060953755	1.029484342	0.511635356	0.10519256
BC0527	TD	1.968893959	0.959778137	0.482383935	0.097721968
BC0660	TD	1.605523829	1.037556564	0.51706284	0.102859965
BC0667	TD	2.009212596	1.005054676	0.500795828	0.093352566
BC0692	TD	2.594427714	0.984343658	0.498997655	0.107982442
BC0695	TD	2.365396878	1.009798866	0.482650165	0.108674079
BC0132	TD	1.955116996	0.959212384	0.514074249	0.104621193
BC0229	TD	1.977207427	1.019388023	0.502095488	0.102472132
BC0261	TD	1.684402081	1.037624994	0.514278807	0.094839458
BC0282	TD	1.871086954	1.041979723	0.501076291	0.105009148
BC0284	TD	1.987015328	0.957214817	0.49212231	0.10530832
BC0296	TD	1.396378641	1.034178433	0.495896867	0.095137366
BC0471	TD	1.697903585	1.047786544	0.515260472	0.105460813
BC0473	TD	1.552476159	0.998215691	0.484903305	0.105648441
BC0492	TD	1.774509503	0.98299658	0.488375275	0.103662385
BC0499	TD	1.823058624	1.005735182	0.491222729	0.100211234
BC0501	TD	1.879745496	1.022199317	0.491731845	0.107233564
BC0563	TD	1.706023741	1.068874893	0.518050062	0.101605732
BC0591	TD	1.811774668	0.978259667	0.483746656	0.099141809
BC0671	TD	1.889028534	1.034527912	0.516708014	0.100296411
BC0702	TD	1.555299442	0.933789811	0.496766408	0.100562749
BC0829	TD	2.38592681	0.985421893	0.490008847	0.090402868
BC0830	TD	1.785477073	0.999636219	0.500268347	0.097789846

BC0841	TD	1.988277579	1.021710043	0.506575285	0.105431893
BC0843	TD	2.243093231	1.068302758	0.492226614	0.092865948
BC0008	TD	1.585532244	1.065863424	0.492658621	0.105222813
BC0020	TD	1.814150132	0.99451898	0.484100568	0.107238994
BC0033	TD	1.794830917	0.93076936	0.487649519	0.098593866
BC0044	TD	2.924488854	0.988224892	0.48490132	0.107803066
BC0073	TD	2.125955874	1.020123598	0.493263285	0.092525317
BC0161	TD	1.758536474	1.059325224	0.513623505	0.105251011
BC0178	TD	2.00082069	1.039780946	0.482220757	0.108468063
BC0181	TD	2.038236724	1.056047016	0.514015455	0.099417504
BC0301	TD	1.445205306	0.954438739	0.494576743	0.109702912
BC0332	TD	1.846788859	0.980822872	0.48136188	0.096095281
BC0494	TD	1.287134864	1.01060958	0.497192956	0.095305983
BC0650	TD	1.597071213	1.010528306	0.501590543	0.094309177
BC0655	TD	1.936814036	1.052235507	0.506686977	0.096608155
BC0680	TD	2.442296323	1.01467333	0.5081776	0.099969678
BC0708	TD	2.476138922	0.940027082	0.495858386	0.105005747
BC0720	TD	1.941648147	1.017223871	0.508257894	0.095315602
BC0726	TD	1.734794495	1.026536238	0.491024686	0.097807866
BC0208	TD	2.005364181	1.012452492	0.492852993	0.107305861
BC0231	TD	2.074477725	0.937788314	0.483357189	0.105632364
BC0418	TD	1.790122462	0.993119679	0.500625918	0.108888642
BC0530	TD	1.986277742	0.95853118	0.501762987	0.096765745
BC0663	TD	1.768509887	0.963336066	0.505366871	0.102068314
BC0696	TD	2.601836724	0.951647689	0.493882787	0.108203735
BC0154	TD	2.21551458	1.06045589	0.48340939	0.100930284
BC0288	TD	1.815378242	0.933410958	0.499509792	0.100836309
BC0325	TD	1.65484153	0.983322843	0.487827915	0.098213415
BC0355	TD	2.259239829	0.97900181	0.505889006	0.101067091
BC0364	TD	1.74086517	1.016825401	0.490524105	0.094733983
BC0427	TD	1.757356614	0.985101615	0.512402728	0.097353348
BC0438	TD	1.3703207	0.989664976	0.518535741	0.095328026
BC0450	TD	1.847357677	1.014783827	0.513094532	0.105746136

BC0452	TD	1.845567452	0.933407081	0.516259811	0.10252046
BC0465	TD	1.974216545	1.040763735	0.509976472	0.103503088
BC0469	TD	2.041916783	1.058145587	0.492713628	0.094500176
BC0474	TD	2.127179862	1.018433234	0.489736128	0.093828841
BC0476	TD	2.284413534	1.01825976	0.507002477	0.09358218
BC0485	TD	1.944074085	1.024165389	0.509423012	0.096834841
BC0574	TD	1.709332877	1.053503738	0.499862023	0.092536934
BC0575	TD	1.899077908	1.020401125	0.515074733	0.102257756
BC0832	TD	1.241068903	0.943629933	0.503392113	0.099037607
BC0839	TD	1.969369711	1.063866227	0.481888991	0.104698583
BC0024	TD	1.853090365	1.007117428	0.504035615	0.093499126
BC0065	TD	2.0313489	0.962557839	0.510671579	0.109094848
BC0157	TD	1.283792367	0.985209562	0.514683903	0.108360406
BC0203	TD	2.263953051	1.043207583	0.499342819	0.104507472
BC0227	TD	2.031311946	1.041410882	0.51364519	0.102678762
BC0334	TD	1.745914882	1.016200815	0.508470321	0.100506839
BC0375	TD	1.594144812	0.960760483	0.483422408	0.094952073
BC0408	TD	1.504232779	0.934323324	0.486810396	0.106536866
BC0653	TD	1.671011793	1.067186135	0.482566289	0.102351102
BC0678	TD	2.569890609	1.026737391	0.501148525	0.092525066
BC0685	TD	2.350896384	1.048705671	0.481907019	0.106055803
BC0713	TD	2.460954295	0.937861859	0.498113297	0.092631835
BC0241	TD	1.756123087	0.944843328	0.517219911	0.10473906
BC0278	TD	1.698701995	0.988220242	0.4846537	0.095084399
BC0416	TD	1.529945007	0.955481817	0.516600876	0.098519529
BC0658	TD	1.856237435	0.969502135	0.490103014	0.09504885
BC0068	TD	1.926620097	0.944586253	0.490562446	0.097682874
BC0103	TD	1.738409491	1.031093865	0.482806203	0.100370722
BC0121	TD	1.785852238	1.035114978	0.502857994	0.101197166
BC0143	TD	1.972937814	1.061087506	0.504040471	0.102375126
BC0151	TD	1.823313727	0.938170553	0.497945379	0.102542066
BC0260	TD	1.725149153	0.93920616	0.519484827	0.105205986
BC0264	TD	1.821765461	1.057170796	0.497426697	0.106281494

BC0324	TD	1.740188957	0.958509195	0.509588974	0.094597085
BC0328	TD	2.084926666	1.053447593	0.510176816	0.10696369
BC0383	TD	2.235114644	1.064184323	0.505539592	0.108117339
BC0384	TD	1.734604669	0.966820021	0.51145255	0.106707031
BC0442	TD	1.760939528	1.019350845	0.484957894	0.097448509
BC0446	TD	1.961180714	1.049343191	0.518452192	0.090890651
BC0497	TD	1.912765112	0.993396004	0.515885644	0.096961822
BC0508	TD	1.77748033	1.056103569	0.505557128	0.103365687
BC0510	TD	2.006017768	0.958360816	0.509813988	0.100166943
BC0582	TD	2.12956149	1.063049785	0.511156494	0.100107766
BC0668	TD	1.721934625	1.034865502	0.482572046	0.099996048
BC0699	TD	2.323058597	0.956601904	0.493890025	0.094385148
BC0705	TD	1.611682985	0.985227407	0.499362321	0.097478378
BC0706	TD	2.406081545	1.005920543	0.514436789	0.097924861
BC0827	TD	1.940085568	0.988307369	0.489966809	0.107358905
BC0833	TD	2.91098483	1.047809072	0.517009493	0.109772333
BC0837	TD	1.955609264	0.932217747	0.486818188	0.102010459
BC0845	TD	2.001475335	0.965886583	0.490114451	0.09882376
BC0005	IVD	2.002453622	0.997106082	0.481502324	0.103373382
BC0018	IVD	1.777934636	0.965100751	0.492494278	0.096694175
BC0026	IVD	1.297506853	1.038303328	0.497479251	0.092364343
BC0031	IVD	1.957679138	1.046403977	0.487188913	0.106572354
BC0046	IVD	1.672867917	1.047215405	0.481089578	0.090946537
BC0047	IVD	1.951436232	0.960072468	0.505344369	0.092651385
BC0053	IVD	1.829859225	1.024050102	0.493026237	0.103144129
BC0057	IVD	1.734064462	0.99091739	0.514936938	0.106552968
BC0072	IVD	2.120505405	1.068557221	0.506291899	0.105523981
BC0099	IVD	1.93527226	1.042484427	0.507188861	0.090958949
BC0109	IVD	2.104283209	0.953827726	0.515079578	0.107266113
BC0179	IVD	2.097026544	1.028204202	0.515329389	0.103885111
BC0209	IVD	2.067974488	0.988863002	0.488630274	0.108668462
BC0212	IVD	2.052004044	1.059551818	0.508143588	0.101253932
BC0215	IVD	2.013168874	1.051547356	0.503232773	0.098899779

BC0222	IVD	1.656126336	0.998132629	0.490151076	0.099714418
BC0224	IVD	1.66987265	1.038863274	0.493798502	0.109879411
BC0239	IVD	2.063198209	1.034537493	0.48366206	0.105523374
BC0298	IVD	1.988773142	1.051918338	0.493820223	0.090244498
BC0323	IVD	1.787926775	0.954362154	0.485145719	0.096194269
BC0330	IVD	1.475075466	0.932126383	0.494151142	0.1073692
BC0340	IVD	1.774545917	0.980284998	0.512418575	0.096368962
BC0342	IVD	1.794956988	1.023281898	0.481008096	0.102964714
BC0347	IVD	2.009069645	1.045764519	0.496571781	0.109173101
BC0348	IVD	1.565329403	1.014472423	0.494153961	0.103907167
BC0350	IVD	1.997960635	1.046022782	0.484852735	0.107543921
BC0356	IVD	1.750945666	1.045861606	0.508986211	0.094473276
BC0362	IVD	1.734234375	0.943051408	0.483550183	0.10700476
BC0363	IVD	1.50868802	1.012820849	0.493282463	0.105664322
BC0365	IVD	1.726328043	1.033563256	0.49223426	0.106981835
BC0371	IVD	1.649722165	0.977687255	0.502874265	0.099558595
BC0372	IVD	1.884899107	1.008672063	0.517725785	0.098624912
BC0376	IVD	1.930085976	1.016896144	0.496606052	0.105881244
BC0379	IVD	1.962769452	0.992446151	0.515387623	0.107185119
BC0381	IVD	1.851430704	1.058463901	0.487107497	0.109894802
BC0392	IVD	1.922233519	1.046524027	0.516962005	0.107418374
BC0397	IVD	1.753920137	0.964329711	0.502639724	0.099724928
BC0399	IVD	1.547567982	1.039523854	0.48816907	0.095543343
BC0402	IVD	1.84229995	0.962298595	0.491583868	0.096257966
BC0412	IVD	1.858699955	1.006531035	0.500981037	0.099953068
BC0523	IVD	2.30988314	0.937900425	0.501316904	0.109037544
BC0764	IVD	2.107487675	1.055815157	0.513705036	0.105147506
BC0765	IVD	1.956726683	1.03603887	0.511778594	0.108273664
BC0766	IVD	2.062366065	1.069810495	0.483140196	0.106491983
BC0767	IVD	1.351422949	1.058697352	0.480026065	0.094395198
BC0112	IVD	1.933928132	0.946623138	0.515433262	0.10141275
BC0141	IVD	1.961127252	1.069007086	0.507825829	0.104012679
BC0170	IVD	1.713047475	0.938129093	0.503927642	0.108548687

BC0234	IVD	1.856740837	0.994842088	0.484191153	0.105004984
BC0240	IVD	2.055047129	1.004207328	0.488145161	0.09905937
BC0242	IVD	1.871047025	1.049033256	0.495537025	0.106269516
BC0254	IVD	1.948005485	1.06602054	0.49574137	0.107431513
BC0255	IVD	1.840419084	0.931204057	0.507895976	0.104488326
BC0289	IVD	1.678930195	0.952699126	0.483683576	0.098656488
BC0319	IVD	2.023870055	1.048489753	0.514309932	0.091548291
BC0351	IVD	1.698595654	1.049117678	0.482152933	0.096571054
BC0385	IVD	1.760624454	0.974449342	0.481866423	0.107725525
BC0396	IVD	1.638824734	0.943086408	0.493710373	0.109491892
BC0401	IVD	1.849822142	1.022350769	0.484871382	0.095312244
BC0409	IVD	1.81380118	0.940628977	0.487888788	0.095657079
BC0414	IVD	1.74006672	1.005045851	0.507173803	0.103293695
BC0426	IVD	1.807082958	1.063056098	0.494504869	0.095488402
BC0432	IVD	1.822600883	1.038531848	0.494244227	0.108321495
BC0433	IVD	1.953495628	1.051267834	0.505210046	0.09425624
BC0435	IVD	1.983230109	1.054835155	0.514765792	0.106969897
BC0445	IVD	2.079924205	1.025449106	0.503080893	0.106778591
BC0484	IVD	1.847905255	1.011292363	0.516355219	0.097562063
BC0525	IVD	1.483604994	1.04668524	0.513476665	0.090255314
BC0528	IVD	1.88272507	0.966292937	0.493446362	0.102482279
BC0536	IVD	1.978634438	1.067171369	0.483357538	0.092125756
BC0540	IVD	1.98876878	1.064021152	0.483903929	0.102767131
BC0541	IVD	1.64755056	0.992812522	0.480369172	0.101764019
BC0543	IVD	1.886380461	1.023171878	0.483760979	0.105900115
BC0544	IVD	1.703479057	1.055690359	0.484042063	0.100791441
BC0547	IVD	1.743263952	1.017752859	0.493234858	0.099538769
BC0549	IVD	1.87294319	1.005322557	0.506307593	0.100686091
BC0551	IVD	1.670264461	1.056561809	0.488010561	0.10331759
BC0555	IVD	1.722626034	1.058120405	0.492373083	0.103936277
BC0556	IVD	1.765292013	0.9988129	0.498261313	0.103226377
BC0557	IVD	2.115450625	1.067027609	0.482366959	0.101159942
BC0560	IVD	1.458655908	1.020841657	0.497048263	0.093052347

BC0564	IVD	1.563737347	0.953417275	0.497061889	0.103358107
BC0566	IVD	1.63185449	1.055778514	0.482295595	0.096678668
BC0568	IVD	1.644320529	0.956683228	0.487872199	0.099132506
BC0570	IVD	1.649453134	0.951062312	0.489407795	0.094382922
BC0587	IVD	1.650329179	1.043711873	0.49278005	0.105305198
BC0728	IVD	1.991573756	0.991233396	0.508802961	0.109628585
BC0729	IVD	1.660096773	1.039685573	0.510729459	0.107752436
BC0730	IVD	1.75628627	1.017927554	0.506335229	0.108389388
BC0731	IVD	1.770970786	0.953500979	0.507171769	0.092862783
BC0732	IVD	1.917272148	0.99279108	0.480482517	0.095011474
BC0733	IVD	1.4716555	0.938184109	0.515346805	0.092519929
BC0734	IVD	2.290783664	1.027511884	0.496729176	0.106142689
BC0735	IVD	1.953785131	1.013652873	0.484791849	0.095951594
BC0736	IVD	2.080092376	1.042255259	0.50163197	0.098215288
BC0737	IVD	2.089946707	0.991967423	0.485186236	0.109313229
BC0738	IVD	2.06375022	1.000676247	0.483728125	0.109607951
BC0739	IVD	2.020147378	1.055412892	0.487299178	0.098882417
BC0740	IVD	2.130472308	0.990435195	0.499268715	0.097087572
BC0741	IVD	1.405461415	1.018236643	0.497255897	0.092483791
BC0742	IVD	2.11202118	0.997768313	0.510576943	0.097374791
BC0743	IVD	1.79392487	0.93129299	0.503173189	0.091604919
BC0084	IVD	1.700610165	1.021940041	0.508412513	0.095511077
BC0089	IVD	1.784924229	1.044751238	0.497333003	0.090617785
BC0116	IVD	1.634956821	1.055770713	0.511636522	0.105056492
BC0124	IVD	1.912765533	1.033276068	0.499658505	0.10397186
BC0136	IVD	1.916504996	1.03892149	0.498745062	0.100103532
BC0140	IVD	1.997819746	1.006395196	0.517124501	0.10272328
BC0144	IVD	1.749505634	1.056615229	0.4856067	0.10900263
BC0147	IVD	1.686457382	0.954631496	0.496722082	0.10895189
BC0163	IVD	3.399401424	1.045964451	0.499736875	0.103941739
BC0166	IVD	1.867278245	0.979429892	0.489486954	0.100318207
BC0217	IVD	1.891048665	0.966721221	0.512685478	0.093935172
BC0223	IVD	1.857174872	1.023626905	0.511270411	0.104426753

BC0226	IVD	1.679183606	1.020239742	0.515191963	0.109937072
BC0235	IVD	1.813970735	1.044617247	0.519545658	0.108263599
BC0244	IVD	2.125162914	1.029282506	0.48089855	0.109146873
BC0267	IVD	1.748366805	1.025652792	0.510486218	0.09641892
BC0268	IVD	1.905247968	1.067029568	0.496464175	0.103615482
BC0270	IVD	1.819351232	1.001808693	0.497086253	0.09691775
BC0272	IVD	1.94894691	1.022874415	0.480372793	0.104856515
BC0280	IVD	1.426435358	1.015138576	0.517725682	0.105681384
BC0283	IVD	1.792121721	1.023864836	0.496158268	0.10780657
BC0290	IVD	1.709729633	1.03702676	0.482299626	0.108914777
BC0291	IVD	1.906965519	1.033130231	0.5193612	0.092197114
BC0297	IVD	1.870655997	1.059692537	0.500176816	0.095317646
BC0300	IVD	1.725277577	1.007353278	0.511677017	0.107470575
BC0303	IVD	2.024200246	0.986680206	0.516971525	0.106915784
BC0307	IVD	1.844262572	1.068279747	0.516101521	0.101811607
BC0308	IVD	1.866550018	1.059079999	0.492164924	0.096860981
BC0310	IVD	1.952310533	0.976444143	0.498313202	0.105294426
BC0312	IVD	1.628235336	0.95004004	0.500153617	0.10909449
BC0313	IVD	2.207016111	0.949879063	0.507517545	0.107756075
BC0315	IVD	1.692441234	1.064183028	0.505045595	0.108635803
BC0321	IVD	1.733407224	1.002533091	0.51984374	0.103541052
BC0326	IVD	1.773454053	1.063816679	0.494123139	0.090141799
BC0331	IVD	1.804182736	1.021528645	0.498320769	0.106336738
BC0339	IVD	2.29534336	1.005501144	0.513238079	0.095799158
BC0357	IVD	1.764339247	0.931524308	0.510465599	0.103409981
BC0370	IVD	2.193934887	0.937836163	0.493011485	0.107205393
BC0373	IVD	2.014327536	1.048592191	0.515255518	0.108836816
BC0377	IVD	2.196305978	1.049764324	0.516690824	0.09146823
BC0386	IVD	2.339389209	1.046969545	0.490682274	0.108152806
BC0391	IVD	1.910388818	1.069283046	0.499592644	0.094851436
BC0394	IVD	2.389006355	0.933848253	0.51349278	0.09775641
BC0407	IVD	1.410734621	1.022908779	0.500531777	0.096817689
BC0413	IVD	1.836434392	1.001356692	0.513782709	0.093057022

BC0417	IVD	1.852876268	0.957202519	0.509546599	0.09971747
BC0421	IVD	2.404822873	1.043664326	0.518491727	0.099812773
BC0434	IVD	1.933416315	0.945342314	0.487603747	0.107953261
BC0439	IVD	2.289492887	1.040224836	0.516183924	0.10449954
BC0441	IVD	1.819799531	0.956075527	0.501920192	0.091463511
BC0443	IVD	1.83418159	1.027590228	0.518643502	0.095475324
BC0448	IVD	1.445486616	1.06720583	0.504782326	0.107384709
BC0449	IVD	1.938265621	1.028823956	0.511658142	0.108072514
BC0451	IVD	1.945830767	1.004217895	0.501903931	0.091445895
BC0456	IVD	2.056316834	1.034775753	0.482227627	0.1067256
BC0458	IVD	1.930696747	1.065806	0.49121775	0.106429888
BC0459	IVD	1.896746426	0.99523883	0.517756609	0.1064941
BC0461	IVD	1.825225552	0.987135479	0.49411274	0.105820482
BC0462	IVD	1.946555912	1.046577306	0.508623748	0.108034857
BC0463	IVD	2.182348541	1.002201081	0.48080136	0.109850354
BC0464	IVD	2.208977078	0.969666003	0.502194567	0.108083113
BC0466	IVD	2.040579025	0.949846568	0.496905937	0.105501413
BC0467	IVD	1.839481506	1.045496504	0.491218071	0.09247714
BC0470	IVD	2.219474086	1.011135746	0.512656475	0.093035993
BC0478	IVD	2.135452259	1.046540218	0.511998541	0.106130787
BC0493	IVD	1.740725642	1.064599618	0.51005249	0.106532642
BC0498	IVD	2.266362372	1.021226754	0.504694328	0.109459134
BC0500	IVD	1.675645203	1.063831527	0.513868383	0.104480995
BC0502	IVD	1.977739045	1.006067264	0.519944708	0.099989448
BC0504	IVD	1.70137085	0.972775261	0.480054461	0.107170213
BC0515	IVD	2.343580395	1.061103855	0.509317669	0.109016875
BC0518	IVD	1.980422329	0.983237559	0.499085104	0.090339892
BC0520	IVD	2.092216792	1.044860971	0.512161992	0.106852516
BC0539	IVD	1.367657158	1.015542329	0.502386839	0.096754521
BC0546	IVD	2.147775516	0.93138762	0.493600316	0.106625046
BC0548	IVD	1.491366621	0.993204023	0.491763526	0.107082152
BC0550	IVD	2.048373134	1.055572109	0.508431043	0.108567466
BC0552	IVD	1.644094388	1.018032996	0.497392172	0.108652015

BC0558	IVD	1.505841255	0.96491763	0.514799491	0.090875589
BC0562	IVD	1.879891598	0.987238058	0.51134327	0.10956054
BC0565	IVD	1.698753537	1.048106562	0.515216457	0.10601107
BC0567	IVD	1.610867148	1.055656246	0.49850269	0.104284732
BC0569	IVD	2.180491082	0.962513972	0.483145379	0.096392885
BC0571	IVD	2.147498441	0.95722062	0.494457656	0.091273662
BC0572	IVD	1.906259826	1.041738659	0.486704612	0.108940732
BC0576	IVD	1.710753906	1.034440097	0.50374377	0.109546981
BC0577	IVD	1.380514197	0.930461941	0.491283363	0.102462992
BC0578	IVD	1.68514306	1.022293784	0.499092825	0.090588422
BC0580	IVD	1.661522669	1.018357898	0.485583014	0.102391994
BC0581	IVD	1.830045321	0.97094413	0.505966776	0.091041214
BC0583	IVD	2.150812033	1.023885383	0.493735643	0.107072146
BC0584	IVD	1.819264214	1.052856313	0.48033345	0.100902742
BC0586	IVD	2.194630927	1.008756236	0.487221865	0.091776266
BC0588	IVD	1.580960011	1.064981003	0.489235048	0.09998067
BC0589	IVD	2.026663841	1.056489529	0.481610584	0.100993185
BC0590	IVD	3.044994654	1.039056376	0.510273507	0.109611441
BC0595	IVD	1.756844045	1.029751928	0.507622389	0.104673491
BC0597	IVD	2.067163026	1.05434973	0.51551802	0.098963515
BC0599	IVD	2.062676507	1.029916375	0.491579614	0.108389214
BC0600	IVD	1.352485111	0.953969999	0.516332507	0.10811527
BC0744	IVD	1.861285927	0.936563102	0.484712963	0.109125117
BC0745	IVD	1.777490146	0.936236376	0.49066687	0.104729928
BC0746	IVD	2.302699749	1.004913091	0.505700109	0.103308901
BC0747	IVD	2.083782325	1.043126107	0.497504586	0.099223514
BC0748	IVD	2.057360678	0.965765035	0.492077588	0.109872307
BC0749	IVD	1.337236818	0.952655896	0.495909182	0.107757633
BC0750	IVD	1.98833434	1.011395128	0.50962407	0.092344428
BC0751	IVD	2.051961593	0.998362985	0.507958714	0.092723275
BC0752	IVD	1.562956727	1.064341316	0.517213068	0.092594094
BC0753	IVD	1.618037359	1.037219666	0.481103109	0.096904027
BC0754	IVD	1.857120159	0.931721952	0.493617183	0.100862752

BC0755	IVD	1.823798976	0.973687149	0.484463017	0.091847743
BC0756	IVD	1.3799353	0.934306345	0.480422081	0.103105153
BC0757	IVD	2.028612644	1.045814591	0.505909661	0.104031617
BC0758	IVD	1.904410354	1.048826891	0.518990192	0.103627941
BC0759	IVD	1.524951511	0.933124283	0.486921058	0.095080496
BC0760	IVD	2.105508218	0.968334171	0.498406698	0.091755962
BC0761	IVD	1.883599139	0.972725479	0.481685374	0.10893715
BC0762	IVD	1.830186066	1.067155632	0.497584021	0.106618632
BC0763	IVD	1.930713246	1.015274694	0.498598972	0.107595956
BC0824	IVD	1.396477373	1.043635939	0.518209058	0.103912751
BC0825	IVD	1.861384523	1.051134825	0.509051575	0.092635643
BC0002	EVD	2.261599915	0.938626608	0.507477484	0.095566475
BC0003	EVD	1.646917546	1.019918532	0.50525963	0.090641034
BC0009	EVD	1.643040809	0.988764812	0.519655564	0.108290389
BC0010	EVD	1.754877417	1.034111517	0.513408265	0.090082532
BC0015	EVD	1.85755293	0.945602809	0.504609788	0.092800333
BC0016	EVD	1.653387261	0.968948612	0.499607672	0.094881229
BC0021	EVD	1.793333267	0.946703286	0.484155278	0.091964134
BC0022	EVD	1.631562951	1.022100289	0.5049166	0.092308599
BC0029	EVD	1.961553019	1.015719866	0.484865023	0.099548502
BC0035	EVD	1.892649136	1.024487807	0.491375562	0.096176541
BC0041	EVD	1.507237183	1.020914875	0.517715315	0.10159839
BC0042	EVD	1.77274334	0.931946764	0.489324361	0.100447174
BC0048	EVD	2.246383659	0.942120872	0.490046973	0.105428521
BC0049	EVD	1.821711987	0.955450738	0.502855803	0.105215464
BC0056	EVD	1.922679508	1.009377001	0.485333533	0.090914503
BC0061	EVD	1.798314845	0.940208849	0.511531898	0.108368554
BC0062	EVD	1.695652613	0.983740218	0.498899411	0.103875114
BC0071	EVD	1.743714645	0.971422971	0.491875203	0.103564821
BC0076	EVD	1.607833776	1.036777555	0.510815521	0.108957007
BC0080	EVD	2.10311736	0.933085699	0.515112234	0.09637841
BC0081	EVD	1.681898516	1.05514939	0.501329918	0.105516426
BC0085	EVD	2.030508977	1.052554638	0.506009954	0.10519014

BC0086	EVD	1.848429663	0.949139624	0.511391709	0.108656294
BC0090	EVD	2.034986605	0.979557181	0.502451493	0.103652297
BC0091	EVD	1.748603832	1.069247179	0.515968685	0.109811633
BC0095	EVD	2.116830675	0.980079502	0.501665114	0.093322689
BC0096	EVD	1.698325019	1.02425296	0.515066489	0.102672244
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BC0110	EVD	1.952527166	1.004451504	0.487201231	0.105914156
BC0111	EVD	1.967512657	0.953487415	0.491099276	0.092010945
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BC0118	EVD	2.092058421	1.047130432	0.503088709	0.096382925
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BC0606	EVD	1.901605464	0.930423416	0.502534893	0.090405235
BC0607	EVD	1.901600618	0.982986044	0.513391246	0.103741085
BC0608	EVD	2.031672924	1.002002586	0.495079895	0.097190604
BC0609	EVD	1.379669373	0.933783393	0.515234857	0.102129394
BC0610	EVD	2.176983351	1.064095206	0.50824889	0.090757505
BC0614	EVD	1.675710577	0.993711669	0.517042931	0.097750151
BC0774	EVD	2.191027028	1.023901293	0.49232269	0.105893419
BC0775	EVD	1.935249178	0.996570343	0.508521784	0.108292791
BC0776	EVD	2.355054231	0.975642412	0.483224331	0.09461017
BC0777	EVD	1.37221724	1.02281587	0.487518589	0.106253175
BC0778	EVD	1.98568167	1.028001142	0.504449489	0.105830375
BC0779	EVD	2.046146876	1.008591786	0.514056276	0.096151348
BC0780	EVD	2.018529807	0.950410435	0.503514587	0.099660677
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BC0784	EVD	2.235156238	1.007860524	0.480749702	0.104005
BC0785	EVD	1.60253501	1.004665125	0.515582989	0.092723217
BC0786	EVD	2.023159404	1.014337743	0.514468379	0.099375338

BC0787	EVD	1.929728722	1.037551418	0.492583535	0.103768999
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BC0791	EVD	2.072250494	1.05808806	0.502259822	0.101769326
BC0792	EVD	2.270769866	1.051494093	0.496657388	0.108673821
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BC0127	EVD	2.387375811	0.950723409	0.508100906	0.102773493
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BC0148	EVD	2.087394226	1.027841027	0.513141163	0.094711217
BC0152	EVD	1.912850129	1.06279455	0.496856419	0.099698935
BC0156	EVD	1.863598226	0.937552999	0.497255308	0.105796767
BC0160	EVD	2.255103061	1.069223015	0.485853995	0.101875749
BC0164	EVD	1.544839885	1.045117348	0.497511809	0.106016914
BC0172	EVD	1.872900386	1.039327822	0.511632318	0.109171707
BC0176	EVD	1.97102639	0.975117528	0.502922536	0.095280042
BC0180	EVD	2.12863921	0.975393783	0.504777944	0.100778014
BC0611	EVD	1.876972242	0.936194048	0.501079856	0.106395261
BC0613	EVD	1.827756676	0.983333514	0.494043278	0.09425843
BC0615	EVD	1.245014728	0.97130742	0.488947464	0.104991731
BC0616	EVD	1.803795765	1.02224801	0.505882438	0.102886552
BC0617	EVD	1.2973166	1.03972355	0.516742631	0.106094293
BC0618	EVD	1.894122614	1.018651006	0.493543118	0.107174351
BC0619	EVD	1.860886773	1.04201899	0.504173796	0.107982525
BC0620	EVD	2.153122288	1.01110854	0.50707364	0.108728692
BC0621	EVD	1.491177299	1.041382212	0.498223834	0.108846386
BC0641	EVD	1.981299413	1.004622257	0.497390726	0.093320554
BC0768	EVD	2.015889425	0.983995612	0.502814829	0.109189104

BC0769	EVD	1.904421518	1.004854879	0.517103135	0.091120513
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BC0796	EVD	2.183763722	1.034741957	0.516649717	0.098931431
BC0797	EVD	2.166169493	1.065539668	0.502138594	0.09019531
BC0798	EVD	1.981632514	1.002458123	0.503393194	0.098692611
BC0799	EVD	1.804131364	0.961337421	0.517580333	0.095956496
BC0800	EVD	1.801836671	1.012858601	0.487976879	0.102674974
BC0801	EVD	1.056544018	0.962351523	0.491941307	0.101162402
BC0802	EVD	1.759606905	1.058578176	0.492571701	0.101625752
BC0803	EVD	1.687689626	1.034892609	0.518181684	0.106274199
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BC0806	EVD	2.197677482	0.982894199	0.497939596	0.105534711
BC0807	EVD	2.103554404	1.040060554	0.496284021	0.090560781
BC0115	EVD	1.968550692	1.025567361	0.515124269	0.090712695
BC0188	EVD	2.066404979	0.980016051	0.509311518	0.100243901
BC0192	EVD	2.021039641	0.973359765	0.486685309	0.090349675
BC0200	EVD	2.162621753	1.032106408	0.509026496	0.104234919
BC0601	EVD	1.903647373	1.036070893	0.51965879	0.109540544
BC0603	EVD	2.291224764	0.977540647	0.493045132	0.109532614
BC0604	EVD	2.061298413	0.997601591	0.519591196	0.105637134
BC0605	EVD	1.91918043	1.00777188	0.497349455	0.109354986
BC0612	EVD	1.967598608	1.043860363	0.493320542	0.101419348
BC0622	EVD	2.011827252	0.982960756	0.50141589	0.10781732
BC0623	EVD	1.938615711	0.983970557	0.513913734	0.107221627
BC0624	EVD	2.029953143	0.938430369	0.516241125	0.103325468
BC0625	EVD	1.880806929	1.022212153	0.51247112	0.097290024
BC0626	EVD	2.189814536	0.973897383	0.488062235	0.105379772
BC0627	EVD	1.785134849	0.966133442	0.488686191	0.097607191
BC0628	EVD	2.002041798	1.041512037	0.506150476	0.100083573
BC0629	EVD	2.066019068	0.955169335	0.500509291	0.101723773
BC0630	EVD	1.933244861	0.933006674	0.503194534	0.109528933
BC0631	EVD	1.856370293	1.017811097	0.503046956	0.107500695

BC0632	EVD	2.196042524	1.033425926	0.519832079	0.104072424
BC0633	EVD	1.499516328	0.990739626	0.486581531	0.104714275
BC0634	EVD	1.848811807	1.036239699	0.502160246	0.102885
BC0635	EVD	1.94424013	1.029717167	0.508191471	0.108480531
BC0636	EVD	1.924225961	0.944264833	0.496694396	0.093853346
BC0637	EVD	1.962172772	0.944496568	0.518028426	0.09312924
BC0638	EVD	1.870275751	1.017399747	0.510931156	0.094355031
BC0639	EVD	1.816368139	1.01839205	0.488822127	0.101297937
BC0640	EVD	2.03098479	0.989575838	0.498465048	0.103936512
BC0642	EVD	1.683844077	1.010511481	0.491379213	0.109860081
BC0643	EVD	2.277656898	1.063080603	0.500935858	0.102394843
BC0644	EVD	1.26117575	1.032928197	0.517675004	0.095659703
BC0645	EVD	1.779893306	0.976373309	0.503533089	0.094613813
BC0646	EVD	1.766115595	1.053325745	0.504120135	0.105177232
BC0647	EVD	1.727039867	0.978140068	0.518085817	0.105983953
BC0770	EVD	1.936373701	1.043662757	0.493345289	0.103683872
BC0771	EVD	1.778203753	1.046525325	0.510463356	0.099042521
BC0772	EVD	1.742484814	1.069603586	0.493007355	0.098300633
BC0773	EVD	1.597833198	0.969607274	0.508114637	0.108470782
BC0808	EVD	2.003561337	1.02787697	0.483241347	0.107944867
BC0809	EVD	1.362643253	0.972050544	0.480007008	0.091507211
BC0810	EVD	1.825645703	1.057011589	0.484824249	0.106524227
BC0811	EVD	1.910659185	0.96247153	0.519584848	0.099693908
BC0812	EVD	1.934407366	0.935316839	0.507035738	0.106631264
BC0813	EVD	1.78223741	0.975733886	0.510318519	0.10757877
BC0814	EVD	2.286297763	1.026251856	0.519059733	0.095403384
BC0815	EVD	1.827166538	0.950248619	0.51271396	0.107684491
BC0816	EVD	1.773895415	1.06092237	0.508208905	0.098077548
BC0817	EVD	1.207462812	1.069240173	0.502372667	0.109544273
BC0818	EVD	1.677384028	0.999326274	0.499807915	0.108606801
BC0819	EVD	1.790378636	0.978374795	0.494211823	0.108015606
BC0820	EVD	1.851616034	0.987798927	0.503465481	0.095546691
BC0821	EVD	1.923167293	0.960766116	0.512742752	0.092514417

BC0822	EVD	2.148624589	1.05299228	0.492750185	0.102417694
BC0823	EVD	1.870609902	1.052053593	0.503944546	0.097346475
BC0128	IMM	1.699180605	0.941504022	0.483793952	0.099285215
BC0204	IMM	2.047290281	0.968151564	0.506987095	0.097207572
BC0404	IMM	2.075547083	1.027019921	0.49739397	0.106417083
BC0424	IMM	1.619987472	0.939492386	0.492243893	0.096343662
BC0158	IMM	1.529092406	1.008360155	0.519382258	0.104692058
BC0014	IMM	1.658739305	1.027654522	0.506293933	0.106277803
BC0104	IMM	1.814390243	1.034911752	0.515982743	0.090024931
BC0195	IMM	1.949313272	1.01682947	0.502757478	0.106264551
BC0206	IMM	1.706682587	0.983127409	0.505186693	0.108993488
BC0275	IMM	1.79635505	1.032172176	0.517471537	0.106823169
BC0295	IMM	2.021638924	1.043387314	0.486084422	0.106542946
BC0322	IMM	1.640782908	0.943452056	0.5123948	0.099171074
BC0366	IMM	1.823037882	0.997213488	0.484943682	0.100923025
BC0060	IMM	1.779607092	0.989130648	0.493382603	0.090098369
BC0130	IMM	1.402158424	1.029350638	0.493785492	0.105485156
BC0233	IMM	1.732945345	1.061528062	0.503172062	0.105689231
BC0259	IMM	1.679477309	0.960964839	0.497532589	0.105377832
BC0131	IMM	1.659045671	1.06224459	0.51738828	0.102923402
BC0142	IMM	1.985765138	1.046125066	0.489366714	0.103809946
BC0184	IMM	2.163973024	1.022266046	0.485888035	0.090623594
BC0134	IMM	1.981076502	0.997229478	0.505911173	0.102345469
BC0346	IMM	1.625235864	0.932120663	0.485593382	0.094120411
BC0382	IMM	1.698627778	1.05315047	0.488371055	0.095414538
BC0538	IMM	1.791519334	0.93703855	0.498932023	0.109117189
BC0602	IMM	1.921701483	0.973398326	0.503118986	0.096788243
BC0285	IMM	1.67523312	0.99651961	0.515282878	0.090432386

RESPONSE TO REVIEWERS' COMMENTS

Reviewer #2 (Remarks to the Author):

The inclusion of details about the model training as well as the feature importance analysis is a clear improvement.

But there are two minor points I still feel need to be addressed:

1) The explanation of the 4 bp end motifs and 6bp breakpoint motifs in the manuscript remains cryptic. The clarification you wrote in the response to my comment is not reflected in the manuscript. In the graphical abstract, the 4bp and 6bp motifs are drawn the same way, and in the text, you do not distinguish between end motifs and breakpoint motifs.

Response: Thank you for the reviewer's suggestion. We have carefully revised the graphical abstract to clarify the difference between the 4-bp end motifs and 6-bp breakpoint motifs, and have also updated the corresponding description in the main text. We appreciate the valuable feedback.

In line 252 you refer to them the same way: "encompassing end motifs (4/6 bp breakpoint signatures)".

Response: Thank you for pointing this out. We have revised the phrasing to "encompassing end motifs (4 bp) and breakpoint motifs (6 bp)" to clearly distinguish between the two. We appreciate your careful reading.

And line 285: "particular 4 bp (e.g., AAGC, AGGC) and 6 bp (e.g., CTTGTT, TGGGAC) end motifs".

Response: Thank you for pointing this out. We have revised the text to "particular 4 bp (e.g., AAGC, AGGC) end motifs and 6 bp (e.g., CTTGTT, TGGGAC) breakpoint motifs" to ensure clarity and distinction.

And line 348: "patterns in breakpoint end motifs".

Response: Thank you for your careful review. We have revised the text to "patterns in end motifs, breakpoint motifs, " to accurately and clearly distinguish between the two concepts.

And line 641: "4 bp and 6 bp end motif frequencies".

Response: Thank you for pointing this out. We have revised the phrasing to "4 bp end motif and 6

bp breakpoint motif frequencies" to maintain clear and consistent distinction between the two terms throughout the manuscript.

2) I am surprised to see that you include so many Repetitive Element features, not just the main classes, but many subclasses. I can understand why Alu elements could be a relevant feature if they are actively transposing in the tumor tissue, but it is only a subset of the Alu subfamilies that can still be active. Furthermore, the results in Supp. Figure 2D shows that the main difference between benign and malignant samples is that there are more Alu reads in a subset of the benign samples. What could explain this? The figure text conflicts with the legend. Line 1065: "breast malignancy (blue) and benign controls (pink)" but the legends specify the colors the other way around. Which is correct?

Response: Thank you for your insightful comment. Our ARTEMIS pipeline analyzes a comprehensive set of repetitive elements, including not only Alu subfamilies but also other classes such as MERs, satellite repeats, LTR elements, and Tigger subfamilies, allowing for a broader feature space.

Regarding the observation that a subset of benign samples shows a higher Alu read count, we agree that this is an interesting finding. One plausible explanation is the natural inter-individual variation in repetitive element content within the general population, where some individuals (and thus some benign samples) may inherently have higher baseline levels of certain repeats. The specific biological mechanisms driving this observation would indeed require further, more targeted investigation. We appreciate your valuable perspective on this point.

We have corrected the description in line 1065 to "cfDNA fragmentomic features between breast malignancy (pink) and benign controls (blue)" to ensure consistency with the figure colors.

Reviewer #2 (Remarks on code availability):

It is not reproducible.

There is no README and no description of the input files (not included). There are some comments in the code but they are all in chinese.

Response: Thank you for your feedback. We sincerely apologize for the oversight. We have updated the code repository to include a comprehensive README file in English. We believe that researchers with basic coding skills will be able to set up and run the code accordingly.

Regarding the full input data, the proprietary, individual patient-level training data cannot be made publicly available due to commercial confidentiality and pending patents. This is clarified in the Data Availability statement. The provided code, example data, and documentation ensure the reproducibility of the analytical methodology.